

Model-based, Learning-based, and Hybrid Whole-Body Control: a Taxonomic Survey of Two Paradigms and Their Convergence

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Abstract

Over the past three decades, whole-body control (WBC) has advanced by piecemeal rather than programmatic progress. Two paradigms developed in near-isolation, a model-based lineage grounded in analytical models and online constrained optimisation, and a learning-based lineage grounded in data and offline optimisation, while a younger current began to fuse them. Existing surveys, however, document only partial technical evolutions separately, and the zone in which the two interpenetrate remains to this day neither delimited nor classified. This survey supplies both map and classification. First, we present the first semantic-bibliometric cartography of WBC: Sentence-BERT embeddings of $n=203$ titles and abstracts, K -means partitioning into tunable K clusters, and UMAP projection render the field’s development as a navigable landscape. Second, we trace each core along an ordering axis introduced here: model-based WBC along a task-arbitration flexibility axis κ_τ , from algebraic null-space priority to predictive whole-body MPC; learning-based WBC along a data-diversity axis κ_d , from reinforcement learning to foundation models. Third, inside this hybrid zone we propose the first topological taxonomy of hybrid model-learning WBC (*Hierarchical, Switching, Parallel, Embedded*), ordered by monotonically increasing coupling between the model cluster $\mathcal{M}$ and the learned cluster $\mathcal{L}$, made operational by a *removability* criterion. Fourth, reading the clusters against one another yields cross-paradigm insights. This work is thus of significant value for understanding the developmental trajectory of WBC, classifying its techniques, and accelerating their convergence.

Keywords: Whole-body control, legged and humanoid robots, hybrid model-learning control, taxonomy, semantic bibliometrics, optimisation-based control

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1 Introduction

Since 2023, commercial redundant robotic systems have been accelerating out of the laboratory into dynamic, unstructured environments that require the execution of multiple tasks. A whole-body controller (WBC) coordinates all degrees of freedom of the robot to realise multiple tasks in parallel, as when tracking an end-effector pose while maintaining balance. WBC is thus the critical enabler for redundant robots entering the real world; this survey focuses on the WBC controller, leaving mechanical design and perception out of scope.

For two decades, methodological development in WBC has been governed by a single dichotomy that partitions the field into two paradigms. *Model-based* methods are formulated upon an explicit mathematical model of the robot and, in exchange, inherit feasibility certificates, contact reasoning, and provable closed-loop behaviour; the price they pay is model accuracy and a brittle response to anything the model has not anticipated. *Learning-based* methods [1], by contrast, can fuse and process raw sensor data and adapt to environmental variation; they incur, however, the risk of unpredictable failure modes and forgo any explicit safety certificate. Within this dichotomy, each paradigm has *abandoned an epistemic resource the other still owns*: model-based methods have surrendered run-time data adaptation; learning-based methods have surrendered explicit physics.

The two paradigms’ weaknesses align, pair by pair. Where model-based methods hit the wall of contact modelling, learning-based methods offer contact *discovery* from rollout data. Where model-based methods struggle with the real-time cost of high-dimensional QPs, learning-based methods compile the same mapping into a single forward pass of a network. Where learning-based methods cannot certify safety, model-based methods offer control-barrier functions whose validity is mathematical. Over the past three decades, WBC papers have piecemeal explored the two paradigms and the *hybrid model-learning* strategies between them. What the field still lacks is a survey that organises these technical points into a coherent, cross-paradigm whole, rather than a scatter of exceptions filed under whichever camp a reviewer favours.

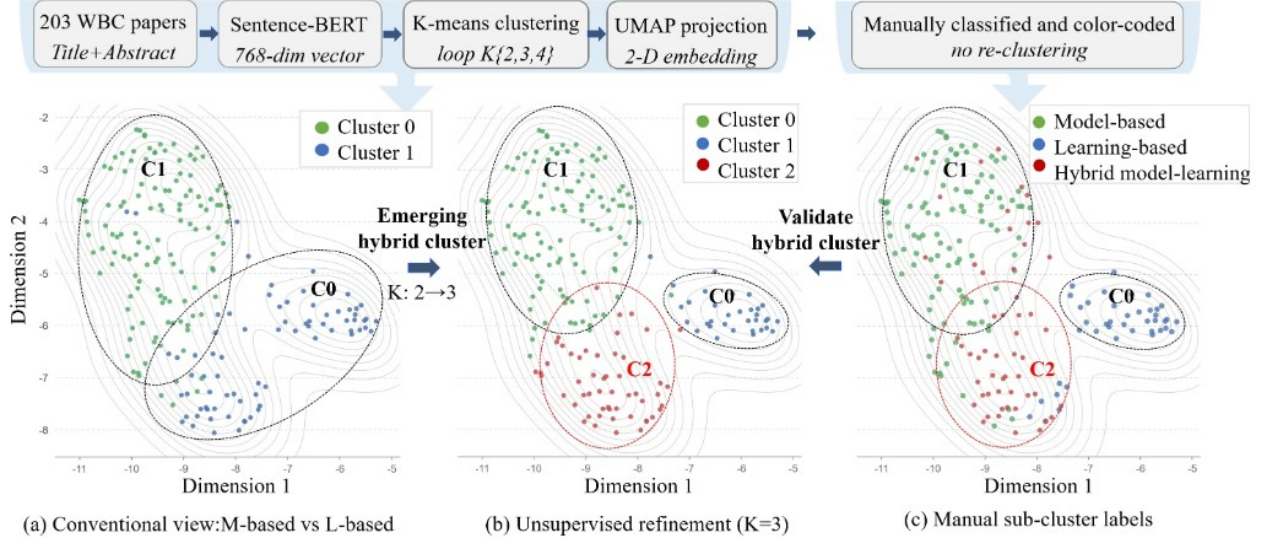


Figure 1: **Semantic bibliometrics of whole-body control (WBC) literature** ($n=203$). Top: Sentence-BERT embeddings are clustered by K -means and projected by UMAP; the same 2-D embedding is then colored by manual labels. (a) With $K=2$, unsupervised clustering recovers a model-side cluster C0 (76) versus a learning-side cluster C1 (127). (b) Refining to $K=3$ keeps a compact model core C0 (33) and a learning core C1 (117), while a hybrid hotspot C2 (53) emerges between them. (c) Manual labels (model-based 41, learning-based 107, hybrid 55) confirm that C2 concentrates the hybrid papers.

A small but growing population of recent papers refuses the binary classification on its face. In Romero *et al.*’s ActorCriticMPC [2] an RL critic is *jointly trained inside* a differentiable MPC and supplies its terminal cost—filing the paper as model-based ignores the critic, filing it as learning-based ignores the MPC backbone. Lin *et al.*’s Residual MPC [3] runs an analytical MPC and a learned residual policy in parallel and adds their outputs at execution time, so neither cluster is auxiliary. Hoeller *et al.*’s Deep Tracking Control [4] stacks the two in a hierarchy: a model-based trajectory optimiser plans contact-rich reference motions that a learned policy then tracks—calling it model-based ignores the policy that delivers the robustness, calling it learning-based ignores the optimiser that defines the task. Each such paper, forced into the binary scheme, would need a fresh footnote.

Yet how do we know this hybrid zone is real, and not an artefact propped up by a handful of scattered examples? Recent surveys of dynamic-legged control [5] acknowledge this difficulty but still defer to authorial editorial judgement. We instead cluster the literature with quantitative tools in a semantic embedding space, and thereby visualise the structure of this hybrid region directly. Sentence-BERT embeddings [6], K -means over a tunable number of clusters K , and UMAP [7] are standard primitives that, in social-science evidence synthesis, have served as the de facto quantitative companion to PRISMA for nearly a decade. We apply them, for the first time, to the epistemic dilemma of WBC.

The pipeline runs in two passes, and only the second pass is the finding. With the number of

clusters forcibly set to $K=2$, unsupervised clustering recovers a model-side cluster C0 (76) versus a learning-side cluster C1 (127) (Fig. 1a). That is, with $K=2$ the method successfully recovers this dichotomy from raw text. Refining to $K=3$ keeps a compact model core C0 (33) and a learning core C1 (117), while a hybrid hotspot C2 (53) emerges between them (Fig. 1b)—a configuration the binary view cannot name. Then comes the cross-check. Working independently of the bibliometric pipeline, and using a four-class taxonomy whose criteria are described in Sec. 5.1, we manually labelled the same $n=203$ papers as model-based (41), learning-based (107), or hybrid (55). The same 2-D embedding, now colored by these labels (Fig. 1c), confirms that C2 concentrates the hybrid papers. Two epistemologically independent paths—one machine-driven over titles and abstracts, one human-driven over full texts—converge on the same region of the embedding space. This convergence, not either pass alone, is what we mean by “the hybrid zone exists.”

Within this hybrid region, as coupling strength and engineering complexity grow together, we divide these works into four ordered classes (Sec. 5.1): HIERARCHICAL, SWITCHING, PARALLEL, and EMBEDDED. These strictly conform to a taxonomy and satisfy a defined membership criterion—the *removability criterion*: peel $\mathcal{L}$ off the deployed system; if $\mathcal{M}$ remains mathematically complete the work is one of the first three classes, and if $\mathcal{M}$ is no longer complete without $\mathcal{L}$ it is EMBEDDED. This monotone coupling axis, moreover, is not peculiar to the hybrid methods: the same organising principle structures our treatment of the model-based methods (ordered

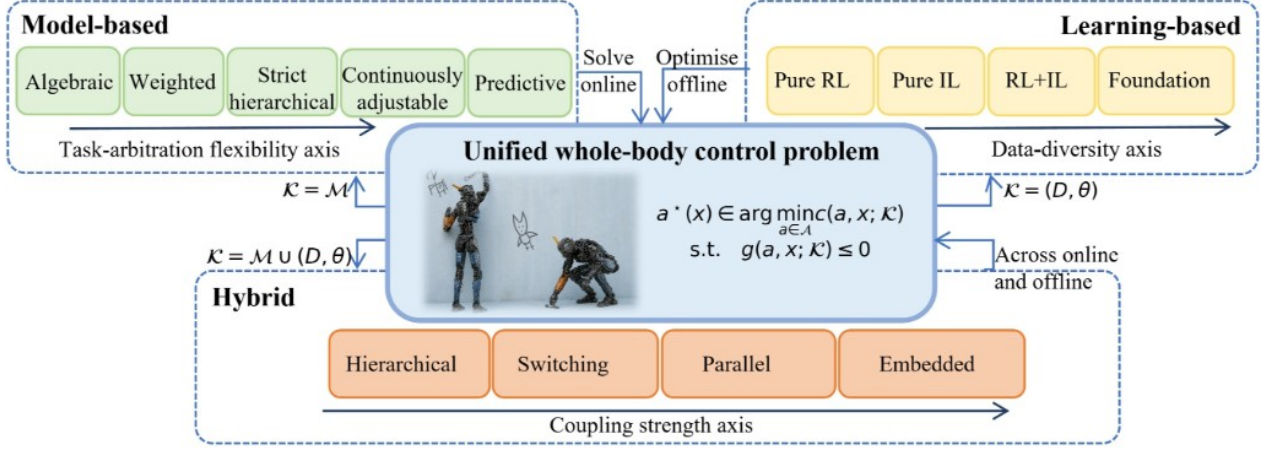


Figure 2: **The three paradigms of whole-body control, each developing along its own coupling-strength axis.** All three paradigms share the same constrained whole-body control problem but differ in the knowledge source $\mathcal{K}$ and in how that knowledge is coupled to the online controller. Model-based WBC orders methods along the task-arbitration flexibility axis κ_{τ} ; learning-based WBC orders methods along the data-diversity axis κ_d ; hybrid WBC orders methods along the model-learning coupling axis κ_{ML} .

by multi-task arbitration) and of the learning-based methods (ordered by the data a policy draws on). The broader insight this affords is drawn out at the close, in Sec. 6.

Contributions and organisation

This survey advances four contributions, each tied to a concrete section of the full document.

1. **A semantic-bibliometrics workflow, applied to WBC for the first time** (Sec. 1): a reproducible pipeline that turns the WBC literature into a measurement rather than a curated viewpoint, whose $K=2$ -to- $K=3$ pass (Fig. 1) dual-validates the empirical existence of the hybrid zone against 55 manually curated papers.
2. **Two sister pillar taxonomies, each ordered by a principled axis introduced here for the first time** (Sec. 3, Sec. 4): model-based WBC ordered by multi-task arbitration (algebraic, weighted, strictly hierarchical, continuously adjustable, predictive), and learning-based WBC ordered by the data prior (pure RL, pure IL, RL+IL, foundation models).
3. **The first systematic taxonomy of hybrid model-learning WBC** (Sec. 5.1): four classes (HIERARCHICAL, SWITCHING, PARALLEL, EMBEDDED) ordered along $\mathcal{M}$ - $\mathcal{L}$ coupling strength, equipped with a membership criterion.
4. **Cross-paradigm insights** (Sec. 6), supported jointly by the three preceding pillars and, by reading them against one another, distilling cross-paradigm insights.

The remainder of the survey applies this organising principle pillar by pillar. The model-based

pillar (Sec. 3) reorganises the WBC canon along a task-arbitration flexibility axis (algebraic, weighted, strictly hierarchical, continuously adjustable, predictive). The learning-based pillar (Sec. 4) is ordered by data diversity (pure RL, pure IL, RL+IL, foundation models). The hybrid zone develops the four-class taxonomy class by class along coupling strength (Sec. 5.2 HIERARCHICAL, Sec. 5.3 SWITCHING, Sec. 5.4 PARALLEL, Sec. 5.5 EMBEDDED). Section 6 then draws the three coupling axes together to yield cross-paradigm insights.

2 Problem Formulation and Notation

Although model-based, learning-based, and hybrid whole-body control appear as an equation, a trained network, or a composition of both, they share the same ingredients: each relies on a knowledge source, each produces a feasible action, and each is composed of a cost that rewards actions and a constraint that decides which actions are feasible. They diverge on only two questions, namely *when* the action is computed and *where* that knowledge comes from.

We now make those shared ingredients concrete. At state x , the controller returns

$$\begin{aligned} a^*(x) \in \arg \min_{a \in \mathcal{A}} c(a, x; \mathcal{K}) \\ \text{s.t. } g(a, x; \mathcal{K}) \leq 0. \end{aligned} \quad (1)$$

where a is the action, x the state, c the task cost, g the feasibility constraint, and $\mathcal{K}$ the knowledge source that feeds both. This is not a new algorithm but a viewpoint that names what every method computes and leaves open how.

We fix notation once and reuse it without restatement. The state is $x = (q, \dot{q}) \in \mathcal{X}$, with q stacking

Table 1: Notation used throughout the survey.

Symbol	Meaning	Domain
$q, \dot{q}$	generalised position / velocity	$\mathbb{R}^{n_q}$
x	state $(q, \dot{q})$	$\mathcal{X}$
a	joint command	$\mathcal{A}$
o	observation	$\mathcal{O}$
r	task reference	$\mathcal{R}$
π	control policy	$\mathcal{O} \times \mathcal{R} \rightarrow \mathcal{A}$
$\mathcal{K}$	knowledge source	
c, g	task cost / feasibility constraint	
$\theta, \mathcal{D}$	policy parameters / dataset	
$\mathcal{M}$	model-based cluster	
$\mathcal{L}$	learning-based cluster	
Φ	composition, $\Phi(\mathcal{M}, \mathcal{L}) = \pi$	
κ_τ	task-arbitration flexibility axis	
κ_d	data-diversity axis	
κ_{ML}	coupling strength axis	

the floating-base coordinates and n_a actuated joints; the action is $a \in \mathcal{A}$, the joint command. An observation $o \in \mathcal{O}$ is whatever the controller reads, and a reference $r \in \mathcal{R}$ carries the task intent, so a policy is the map $\pi : \mathcal{O} \times \mathcal{R} \rightarrow \mathcal{A}$. We write $\mathcal{M}$ for a model-based cluster, $\mathcal{L}$ for a learning-based one, and Φ for the composition that assembles them. Table 1 collects these symbols.

2.1 The Model-based Interface

Model-based control instantiates the unified problem by solving it online with $\mathcal{K}$ drawn entirely from an analytical model. The knowledge source is the model and the timing is online, so (1) is solved afresh once per control cycle on the deployed robot [5, 8]. The feasibility constraint g is formulated from rigid-body dynamics, contact conditions, and friction cones, and the cost c from explicit task objectives. Its strength and its fragility share one root, since the controller trusts g to be correct, so it is exact when the model holds and brittle the moment it does not.

Within this paradigm, methods are ordered by how flexibly simultaneous tasks are arbitrated, namely the task-arbitration flexibility axis κ_τ . The spectrum widens in five steps. Algebraic null-space projection constructs the priority of adjacent equality tasks. Weighted optimisation blends inequality tasks softly into a single cost. Strict hierarchical optimisation enforces the arbitration of multiple tasks [9]. Continuously adjustable priority then makes the ordering itself a control variable, letting tasks trade rank smoothly at run time. Whole-body predictive control finally lifts arbitration into a prediction horizon, resolving conflicts over time rather than instant by instant [10]. Each step buys more flexible arbitration at a higher computational price.

2.2 The Learning-based Interface

Learning-based control solves the same problem of-line, compiling $\mathcal{K}$ from sampled data into a feed-forward policy. The knowledge source is data and the timing inverts, so the optimisation runs offline during training, and at run time the controller is a single forward pass [1, 11]. The feasibility constraint g is no longer explicit; it is approximated implicitly, encoded in a reward or a dataset and absorbed into the parameters θ . Generalisation then follows the coverage of the data, and verifiability fades as the explicit g disappears.

Within this paradigm, methods are ordered by the diversity of the data prior the policy draws on, namely the data-diversity axis κ_d . What grows along the axis is the scale and heterogeneity of that prior, not the rigidity with which any one dataset is reproduced. Pure reinforcement learning draws on the rollouts its own reward generates. Pure imitation adds an external corpus of expert demonstrations. Their combination, the current mainstream, widens that corpus and pairs it with reward-driven interaction [12]. It therefore rests on a more heterogeneous prior than either parent, even though each single demonstration binds it more loosely. Foundation-model policies sit at the far end, drawing on web-scale multimodal corpora rather than a task-specific set.

2.3 The Hybrid Interface

Hybrid control uses both paradigms at once, and the optimisation is split across offline and online stages. It therefore takes an intermediate value on each question rather than committing to an extreme. The structural form of the hybrid determines the four structures catalogued in Sec. 5.

These structures order the family by the coupling strength κ_{ML} between $\mathcal{M}$ and $\mathcal{L}$. The spectrum tightens from *hierarchical*, where a narrow unidirectional set-point links the two, through *switching*, where a discrete signal selects one complete controller at a time, and *parallel*, where both emit actions that are summed, to *embedded*, where they share one computation graph. The boundary that matters most, between a model merely tuned by learning and one fused with it, has a sharp test.

Definition 2.1 (Removability). *Detach $\mathcal{L}$ from a deployed policy $\pi = \Phi(\mathcal{M}, \mathcal{L})$. If $\mathcal{M}$ remains mathematically complete and runnable on its own, the composition is hierarchical, switching, or parallel; if $\mathcal{M}$ is no longer complete without $\mathcal{L}$, because $\mathcal{L}$ supplies the cost, the dynamics, or a differentiable layer, the composition is embedded.*

We begin with the pure model-based end and follow κ_τ (Sec. 3), then the data-diversity axis κ_d (Sec. 4) and the coupling axis κ_{ML} between $\mathcal{M}$ and $\mathcal{L}$ (Sec. 5); Sec. 6 brings the three axes together.

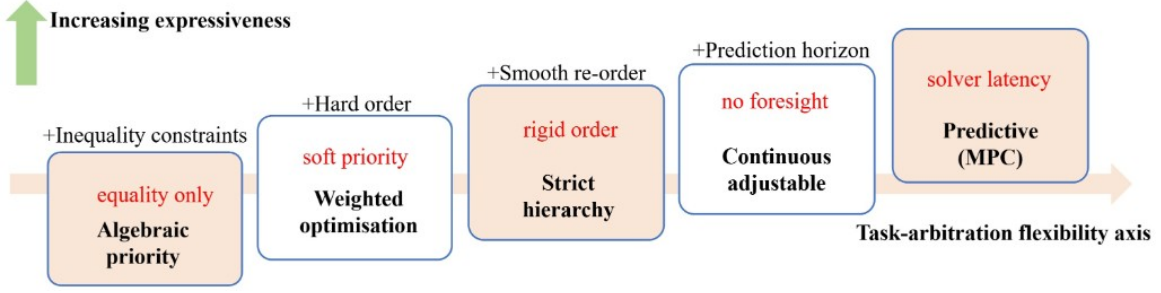


Figure 3: The model-based task-arbitration flexibility axis κ_τ . Moving from algebraic priority to whole-body MPC, each stage resolves the limitation exposed by the previous one: equality-only projection motivates inequality constraints; soft priority motivates hard ordering; rigid order motivates continuously adjustable re-ordering; lack of foresight motivates predictive arbitration. The vertical direction indicates increasing expressiveness of $\mathcal{A}_{\kappa_\tau}$, while the horizontal direction follows the progression of task-arbitration mechanisms.

3 Model-based Whole-body Control

Within the unified formulation of Sec. 2, model-based whole-body control specialises the knowledge source $\mathcal{K}$ to an analytical model and solves (1) online at deployment [5, 13]. Its strength and its fragility share one root: when the dynamics model, the contact model, and the task specification are accurate, the controller computes a feasible joint command every cycle without ever having learned from data; when the model is misspecified, the resulting commands inherit that error and cannot recover information the model was never written to represent.

The map is grounded in multi-contact dynamics, which we write explicitly here for the first time:

$$M(q)\ddot{q} + h(q, \dot{q}) = S^\top \tau + J_c(q)^\top \lambda, \quad (2)$$

where q stacks the base configuration and n_a actuated joints, M is the inertia matrix, h collects Coriolis, centrifugal, and gravity terms, S selects the actuated joints, J_c is the contact Jacobian, λ the contact wrench, and τ the joint torque [13, 14]. A model-based WBC instance forms task residuals $e_i(q, \dot{q}, \ddot{q}, \lambda, \tau)$ and solves, at each control cycle, an online constrained program

$$\begin{aligned} \min_{\ddot{q}, \lambda, \tau} \quad & \mathcal{A}_{\kappa_\tau} \{e_1, \dots, e_N\} \\ \text{s.t.} \quad & M(q)\ddot{q} + h(q, \dot{q}) = S^\top \tau + J_c(q)^\top \lambda, \\ & J_c \ddot{q} + \dot{J}_c \dot{q} = 0, \\ & C_{\text{fric}} \lambda \leq 0, \\ & \tau_{\min} \leq \tau \leq \tau_{\max}. \end{aligned} \quad (3)$$

in which the equality reproduces (2), the second line keeps the feet on their contacts, and the inequalities enforce the friction cone and actuation limits. Everything specific to a method is hidden in the arbitration operator $\mathcal{A}_{\kappa_\tau}$. Read in this light, two decades of model-based WBC answer one recurring question with progressively more capable robot systems: *how should simultaneous whole-body tasks be arbitrated when they compete for the same degrees of freedom?*

The axis κ_τ introduced in Sec. 2.1 orders the paradigm by *arbitration capability*: how rich a set of constraints it can honour, and over how long a horizon it reasons. Five sampling points along that axis structure the section (Fig. 3): algebraic priority, weighted optimisation, strict hierarchy, continuously adjustable priority, and predictive arbitration. The progression is not monotone in a single quantity; rather, each stage remedies a well-defined inadequacy of the preceding formulation and, in extending arbitration capability, thereby surfaces the next binding constraint on task arbitration.

3.1 Algebraic Priority: Null-Space Projection

The earliest answer to task conflict was algebraic. For two velocity-level tasks $J_1(q)\dot{q} = \dot{x}_1$ and $J_2(q)\dot{q} = \dot{x}_2$, null-space projection solves the high-priority task first and confines the second to the directions the first leaves free [15, 16]. A compact two-task solution is

$$\begin{aligned} \dot{q} = & J_1^\# \dot{x}_1 + (I - J_1^\# J_1) \tilde{J}_2^\# (\dot{x}_2 - J_2 J_1^\# \dot{x}_1) \\ & + (I - J_1^\# J_1) (I - \tilde{J}_2^\# \tilde{J}_2) z, \end{aligned} \quad (4)$$

with $\tilde{J}_2 = J_2(I - J_1^\# J_1)$ and z free in the remaining null space. The construction extends recursively to an arbitrary stack,

$$\begin{aligned} \dot{q}_i = & \dot{q}_{i-1} + (J_i N_{i-1})^\# (\dot{x}_i - J_i \dot{q}_{i-1}), \\ N_i = & N_{i-1} (I - (J_i N_{i-1})^\# J_i N_{i-1}). \end{aligned} \quad (5)$$

the algebraic core of the Siciliano and Chialverini task-priority line [17, 18]. This is the first point on κ_τ : priority is *exact*, because it is built into the projection operator rather than negotiated by a solver.

Around this core the early literature added robustness and reach. Damped least-squares pseudo-inverses tame the singular configurations where naive inversion explodes [19, 20], and a comparison of the Siciliano and Chialverini schemes showed that damping the augmented-Jacobian projection better

Table 2: Summary of multi-task arbitration methods for whole-body control.

	Main model	Priority	Advantages	Disadvantages
Algebraic projection [17]	$\dot{q}_i = \dot{q}_{i-1} + \tilde{J}_i^\# (\dot{x}_i - J_i \dot{q}_{i-1})$	strict hierarchy	-simple -low computational -highly efficient	-no inequality constraints -no priority changes
wQP [30]	$\min_{V_{wb}} \frac{1}{2} \ V_{wb}\ _W$	adjustable weight value	-high real-time performance -single QP -low computing power	-no priority changes -low accuracy
HQP [33]	$\min_{x, w_2} \ w_2\ ^2$	strict hierarchy	-resolve hierarchical issues of iQP	-high computing power -high accuracy demands multiple QP iterations
CNP [24]	$x = A_1^\# b_1 + (A_2(I - V_1 S_1^\# S_1 V_1^\#))^\# (b_2 - A_2 x_1) + \dots$	dynamic hierarchy	-smoothness guarantee	-no inequality constraints -cannot alter the global priority order
LIPT [38]	$\dot{q} = \beta(t) \dot{q}_{[T_1 > T_2]} + (1 - \beta(t)) \dot{q}_{[T_2 > T_1]}$	dynamic hierarchy	-faster than layer-by-layer QP -no need to rebuild null space	-cannot enforce new lexicographic order -no inequality constraints
IDVA [41]	$\hat{x}_i = \beta(t) \dot{x}_{di} + (1 - \beta(t)) J_i \dot{q}_{p/i}$	dynamic hierarchy	-simple -adjust any number of tasks	-computational load increases significantly with the number of tasks -no inequality constraints
HQP with continuous priority transition [42]	$\min_{\dot{q}_k, A_{w_k}, B_{w_{k-1}}} \frac{1}{2} \left\ \frac{\beta_A(t) (A_{w_k})}{\beta_B(t) (B_{w_{k-1}})} \right\ ^2$	dynamic hierarchy	-inequality constraints	-transition to adjacent-swap only
GHC [45]	$\arg \min_{\tilde{q}, x} \sum_{i=1}^{n_x} \left\ f_i(\tilde{q}_i, \tilde{\zeta}_i^d) \right\ ^2 + \left\ \tilde{q}' \right\ _{\chi}^2$	dynamic hierarchy	-always a single QP problem -adjust any number of tasks	-low-priority tasks may experience null-space drift
MPC [10]	$\min_{\{x_k, u_k, \lambda_k\}_{k=0}^{H-1}} \sum_{k=0}^{H-1} \ell_k(x_k, u_k, \lambda_k) + \ell_H(x_H)$	predictive hierarchy	-temporal foresight -contact scheduling -real-time reduced models	-lose joint dynamics -need subordinate WBC -high computational cost

suppresses singularity-induced fluctuations, at the cost of biasing low-priority solutions [21]. The same projection idea was carried onto humanoid platforms (HRP-2, ASIMO, Justin, and Hume) to deliver collision-free whole-body motion [22–25], and Sentis’ operational-space stack nested constraint, posture, and manipulation behaviours inside a single hierarchy of torque commands [26]. The limitation is structural rather than incidental: algebraic projection handles equality tasks gracefully, but unilateral contact, friction cones, centre-of-pressure bounds, torque saturation, and collision margins are all *inequalities*. Once these become decisive, arbitration must move from closed-form projection into optimisation.

3.2 Weighted Optimisation: Soft Arbitration

Casting WBC as a numerical program lifts exactly that restriction: it was shown early that algebraic WBC can be rewritten as an optimisation problem, after which inequality constraints enter naturally [27]. The simplest such form is weighted quadratic programming (wQP), which collapses all task residuals into one weighted least-squares objective,

$$\begin{aligned}
 \min_{\mathbf{z}} \quad & \frac{1}{2} \sum_{i=1}^N \|A_i \mathbf{z} - b_i\|_{W_i}^2 \\
 \text{s.t.} \quad & M\ddot{q} + h = S^\top \tau + J^\top \lambda, \\
 & C_{\text{fric}} \lambda \leq 0, \\
 & \tau_{\min} \leq \tau \leq \tau_{\max}.
 \end{aligned} \tag{6}$$

with decision vector $\mathbf{z} = [\ddot{q}^\top, \lambda^\top, \tau^\top]^\top$ and task weights W_i . This is the second point on κ_τ : tasks

are no longer separated by null spaces but blended inside one objective that can simultaneously carry equalities, inequalities, contact forces, and actuation limits.

Operational-space wQP was first brought to free-floating humanoids by prioritising tasks within the operational-space formulation [28], and enhanced OS-WBC variants enforce an approximate order by scaling the diagonal task weights, so that higher tasks dominate [29]. The same formulation has driven mobile collaborative manipulators [30], attractor-based whole-body motion [31], and series-elastic point-foot biped walking [32]. Its convenience is also its weakness: a large weight only *behaves* like a priority. Under strong conflict or near infeasibility, a high-priority objective can still be traded against lower ones, because the ordering is numerical rather than hard. Weighted QP thus widens the constraint vocabulary but weakens the meaning of hierarchy.

3.3 Strict Hierarchical Optimisation: Hard Arbitration

Hierarchical quadratic programming (HQP) recovers hard priority while keeping the inequality constraints that wQP made possible. It solves a cascade in which each level may use only the freedom that leaves every higher level at its optimum. A two-level form makes the idea explicit,

$$\begin{aligned}
 \min_{x, w_1} \quad & \|w_1\|^2 \quad \text{s.t.} \quad A_1 x \leq b_1 + w_1, \\
 \min_{x, w_2} \quad & \|w_2\|^2 \quad \text{s.t.} \quad A_1 x \leq b_1 + w_1^*, \\
 & A_2 x \leq b_2 + w_2.
 \end{aligned} \tag{7}$$

Table 3: The five stages of model-based whole-body control along the task-arbitration flexibility axis κ_τ .

Stage	Solve form	Ineq.	Priority semantics	Temporal scope	Per-cycle cost	Typical solver	Representative methods
Algebraic	closed-form projection	no	exact (by construction)	instantaneous	lowest (one pass)	pseudo-inverse / SVD	null-space, Siciliano, Chiaverini
Weighted	single QP	yes	soft (weights)	instantaneous	one QP	dense QP	OS-WBC, weighted QP
Strict	cascaded QP	yes	hard	instantaneous	one QP per level	cascaded / null-space QP	HQP, inverse-dynamics HQP
Continuous	morphing QP/projection	yes	continuously adjustable	instantaneous	one-few QP per cycle	QP + projector blend	CNP, IDVA, GHC/DynGHC
Predictive	horizon OCP	yes	temporal	horizon H	OCP over H steps	(N)LP/QP, DDP/SQP	centroidal/lumped-mass whole-body MPC

where w_1^* freezes the first level’s optimal slack. The construction generalises to level $k+1$ by optimising the new task only over directions that preserve the residuals already fixed at levels $1, \dots, k$, the framework that generalised task priority to inequality tasks [33]. Fast online formulations make the cascade tractable for humanoid motion generation [27], null-space reformulations cut the constraints carried into each level [34], and operational-space inverse-dynamics solvers extend it to rigid contacts and other unilateral constraints [35].

This is the third point on κ_τ , and the first to give priority a genuine mathematical semantics: a low-priority task may fail, but it may never degrade a higher-priority optimum. That guarantee made HQP a natural fit for humanoid balance and contact-rich control, demonstrated on HRP-2 and on the torque-controlled Sarcos humanoid via momentum-based hierarchical inverse dynamics [33, 36]. The cost is rigidity and latency: the order must be declared in advance, changing it abruptly can produce discontinuous velocities, accelerations, or torques. The next stage therefore asks not how to make a hierarchy strict, but how to let it *change* smoothly.

3.4 Continuously Adjustable Priority: Time-Varying Arbitration

Real tasks rarely keep a fixed order. The right priority shifts with the contact phase, the distance to an obstacle, the support polygon, or the freedom momentarily available. A classical trigger is reachability: when a collision-avoidance torque τ_0 lies mostly outside the null space of the nominal task, a criterion such as

$$c = \frac{\|(I - J^\top J^\top \#) \tau_0\|}{\|\tau_0\|} \quad (8)$$

signals that its priority should rise, and the command is blended through a scalar gate $f(\alpha)$,

$$\tau = f(\alpha) J^\top F + f(\alpha) (I - J^\top J^\top \#) \tau_0 + (1 - f(\alpha)) \tau_0. \quad (9)$$

so the avoidance task is inserted or withdrawn continuously [37]. The general difficulty is not activation but *continuity*: discrete re-scheduling of a strict hierarchy causes jumps in the commanded rates and torques [38].

Continuous null-space projection drives the singular values of a projection matrix from one to zero, smoothly releasing or occupying directions without

rewriting the hierarchy [24, 39, 40]. *Solution morphing* interpolates between the outputs or the desired values of two priority structures (the linear-interpolation transition that motivates this family [38], and the intermediate-desired-value approach that moves the interpolation into task space [41]), and HQP-based morphing extends it to inequality constraints and non-adjacent swaps at the price of extra QP solves [42–44]. *Generalized hierarchical control* (GHC) encodes pairwise task influence in a priority matrix ψ and realises any ordering through a generalized projector solved by a single QP [45, 46], which DynGHC then endows with dynamic consistency [47]. Continuously adjustable priority is the fourth point on κ_τ : arbitration becomes a time-varying object rather than a fixed order. Yet it still decides *instant by instant*, with no view of the future beyond the current cycle.

3.5 Whole-Body Predictive Control: Predictive Arbitration

Continuously adjustable priority still arbitrates instantaneously; whole-body model predictive control (MPC) lifts arbitration into a prediction horizon. Instead of resolving the best compromise now, it optimises a trajectory of states, contacts, forces, and commands,

$$\begin{aligned} \min_{\{x_k, u_k, \lambda_k\}_{k=0}^{H-1}} & \sum_{k=0}^{H-1} \ell_k(x_k, u_k, \lambda_k) + \ell_H(x_H) \\ \text{s.t.} & \quad x_{k+1} = f(x_k, u_k, \lambda_k), \\ & \quad \lambda_k \in \mathcal{C}_{\text{fric}}, \\ & \quad u_k \in \mathcal{U}. \end{aligned} \quad (10)$$

and applies only the first command before re-solving. Arbitration is now *temporal*: a task may be relaxed at the present instant because the horizon predicts recovery, or a contact chosen because it improves feasibility several steps ahead. This is why MPC became central to dynamic legged locomotion, where contact timing, centre-of-mass motion, and force limits must be coordinated over time [5, 10].

The practical art is choosing a model fast enough to solve yet rich enough to predict what matters. Reduced-order centroidal and lumped-mass models keep convex MPC real-time on highly dynamic quadrupeds such as MIT Cheetah 3 [10, 48]; full optimisation-based planning, estimation, and control was demonstrated on the Atlas humanoid under severe contact changes [49]; and online nonlinear motion optimisation extends the same predictive

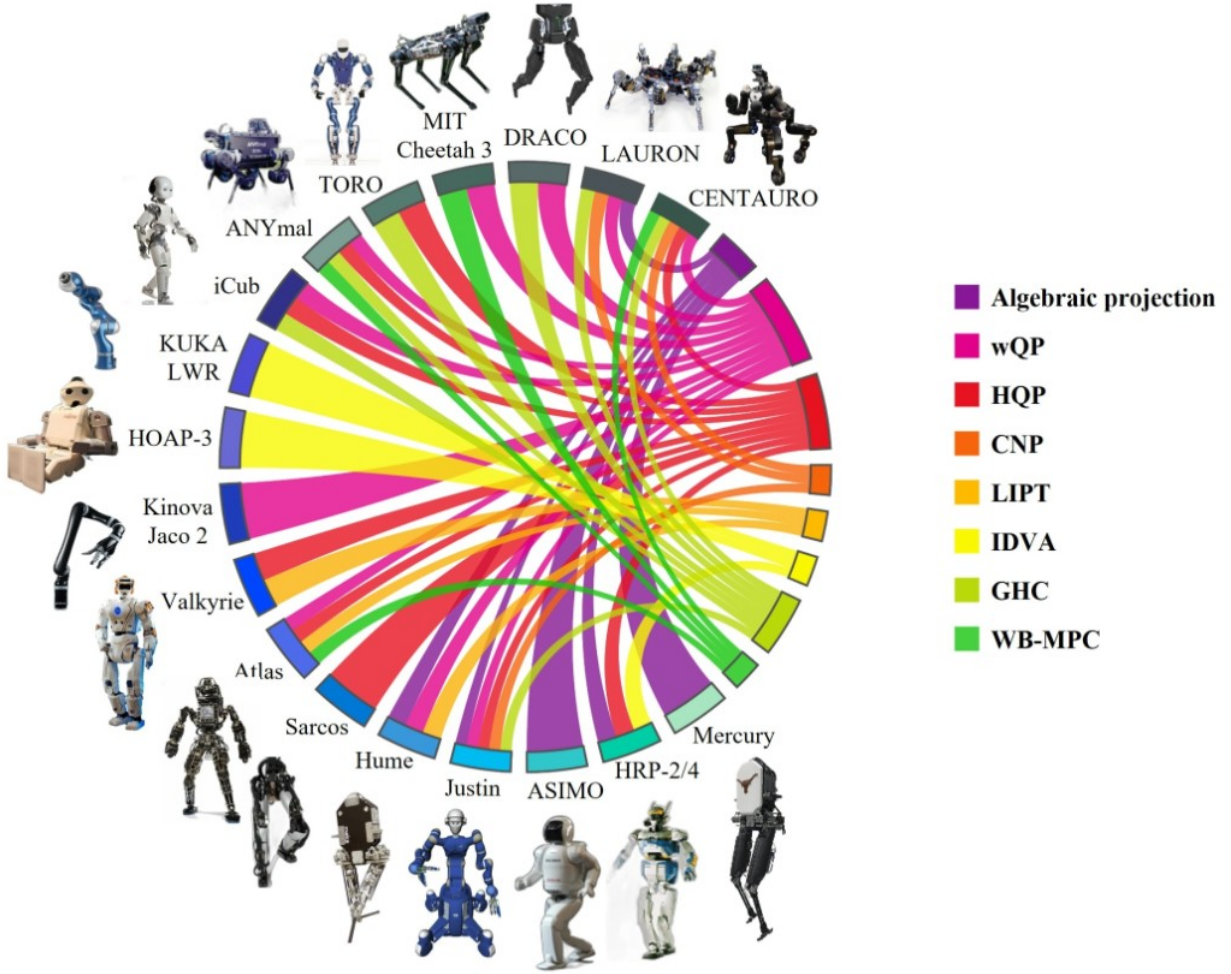


Figure 4: Platform–method map along κ_τ ; band width encodes how prominently each platform employs a given arbitration mechanism.

logic to quadrupedal and wheeled-legged platforms [50, 51]. Predictive arbitration is thus the endpoint of κ_τ : it spends computation to buy foresight, resolving conflicts across time that every earlier stage could only resolve in the instant.

3.6 Structural Bottlenecks

Two structural bottlenecks bound the model-based paradigm, and together they motivate the two pillars that follow.

The first is *solver latency*. Advancing along κ_τ —from algebraic projection to wQP, HQP, continuously adjustable priority, and MPC—steadily enlarges $\mathcal{A}_{\kappa_\tau}$, but every step also adds decision variables, active constraints, horizon length, or repeated QP solves. A controller that misses its real-time deadline forfeits the very online guarantee that justified the analytical model in the first place.

The second is the *model-fidelity ceiling*. Every method in this section trusts the analytical feasibility constraint: contacts are known or predicted, friction cones are meaningful, inertial parameters are accurate, and unmodelled compliance or terrain de-

formation is small enough for feedback to absorb. When the world violates these assumptions, tighter arbitration does not help—it only solves the wrong problem more precisely.

4 Learning-based Whole-body Control

The preceding chapter approached humanoid whole-body control through explicit models of dynamics, contacts, and task constraints. This chapter turns to learning-based control, whose role has grown with advances in large-scale simulation and neural policy optimisation. Rather than modelling every contact mode and behaviour analytically, learning-based methods acquire control mappings from interaction or demonstrations. This offers a flexible route to agile, multifarious, and expressive behaviours, as well as adaptation across motions, environments, and embodiments. The trade-off is greater dependence on data coverage, reward design, and the training distribution.

The two main routes are reinforcement learning

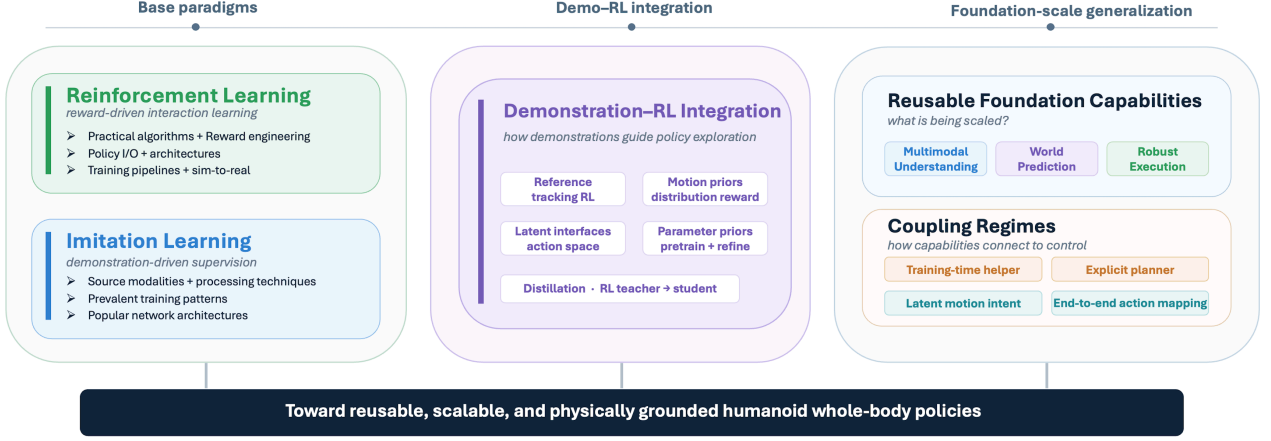


Figure 5: Overall framework of learning-based humanoid WBC.

(RL) and imitation learning (IL). RL learns through environment interaction and task rewards, but the high-dimensional, discontinuous, and balance-critical dynamics of humanoid WBC make unconstrained exploration costly and unstable. IL narrows the motion space by imitating expert demonstrations, but remains vulnerable to covariate shift, embodiment mismatch, and sim-to-real gaps. Successful systems therefore often combine these two routes, where imitation provides a warm start, while RL improves physical feasibility, robustness, and task performance [52–54]. More recently, behaviour and multimodal foundation models have extended this pattern through large-scale pretraining, aiming to turn broad motor competence and semantic understanding into reusable whole-body controllers [55–58]. The following content of this chapter will sequentially visit these concepts.

4.1 Reinforcement Learning

4.1.1 Practical RL Regimes for Humanoid WBC

Reinforcement learning acquires closed-loop humanoid control through environment interaction and task rewards. In humanoid WBC, high-dimensional actuation, underactuated balance, discontinuous contacts, and costly failures make training tightly coupled to simulation scale, and most practical humanoid systems therefore learn online in massively parallel simulation [59, 60], which makes fresh roll-outs inexpensive, leading to stable on-policy optimisation a natural fit for this setting despite its limited sample reuse. PPO has consequently become the established training backbone for humanoid WBC [61]. Its conservative policy updates integrate reliably with privileged critics, curricula, and domain randomisation, forming a mature and scalable training stack.

Recent results, however, challenge the assumption that this on-policy recipe is the only scalable choice. Off-policy algorithms such as FastTD3 and

Table 4: Common reward and termination terms in RL-based humanoid whole-body control. Canonical forms are shown; practical implementations typically apply task-specific weights, clipping, or exponential kernels.

Category	Representative terms and canonical forms	
Task / interaction	Base velocity	$-\ v_{xy} - v_{xy}^*\ _2^2$
	Heading	$- \omega_z - \omega_z^* ^2$ or $-\text{ang}(\psi, \psi^*)^2$
	Task-space target	$-\ x_{ee/\text{obj}} - x_{ee/\text{obj}}^*\ _2^2$
	Force / wrench target	$-\ w_{ee} - w_{ee}^*\ _2^2$
	Reference motion	$-\ q - q^{\text{ref}}\ _2^2 - \ \dot{q} - \dot{q}^{\text{ref}}\ _2^2$
Stability / safety	Uprightness	$-\ g_{\text{proj}} - g_0\ _2^2$
	Body height	$- h - h_0 ^2$
	Fall termination	$-\mathbb{I}_{\text{fall}}$
	Undesired contact	$-\sum_j \mathbb{I}(c_j^{\text{bad}})$
Contact / gait	Foot slip	$-\sum_i c_i \ v_{\text{foot},xy}^i\ _2^2$
	Foot placement	$-\sum_i \ p_{\text{foot}}^i - p_{\text{foot}}^{i,*}\ _2^2$
	Contact schedule	$-\sum_i c_i - c_i^* $
	Foot clearance	$-\sum_i (1 - c_i) [h_{\text{clr}} - h_{\text{foot}}^i]_+^2$
Regularisation	Torque / power	$-\ \tau\ _2^2 - \tau^\top \dot{q} $
	Action smoothness	$-\ a_t - a_{t-1}\ _2^2$
	Joint smoothness	$-\ \dot{q}\ _2^2 - \ \ddot{q}\ _2^2$
	Posture / limits	$-\ q - q_0\ _2^2 - \phi(q)$

FastSAC combine replay with massive parallelism to train high-dimensional humanoid controllers efficiently, while FlashSAC explicitly controls critic error amplification under bootstrapped value learning [60, 62, 63]. These advances show that replay-based RL can match or exceed on-policy training efficiency when critic stability, data coverage, and update scale are carefully controlled.

Offline RL removes corrective interaction entirely. A fixed dataset cannot repair missing state–action coverage, making distribution shift especially restrictive for balance- and contact-critical control. Consequently, pure offline RL remains less established as a deployment recipe for physical humanoid WBC than simulation-first online learning.

Table 5: Common observation and action interfaces in RL-based humanoid whole-body control.

Interface	Typical signals
Observation interface	
Proprioception	Joint / base state; projected gravity; previous action
Task / motion conditioning	Velocity or gait command; task goal; motion reference
External perception	Terrain / depth features; visual or scene representation
Temporal / state inference	Observation-action history; latent dynamics; reconstructed state
Privileged training	Full simulator state; terrain / contact truth; dynamics parameters
Action interface	
Joint-level command	Position / offset target; gain or impedance modulation; torque
Task / planning target	End-effector or footstep target; gait command; MPC / reduced-order parameter
Residual command	Joint / torque correction over a reference or nominal controller

4.1.2 Reward Engineering for Humanoid WBC

Reward engineering maps task intent onto the balance- and contact-critical dynamics of humanoid control. As summarised in Table 4, practical objectives typically combine task tracking, stability and contact shaping, and actuation regularisation. Locomotion policies track quantities such as base velocity and heading, while terrain-aware and locomanipulation tasks additionally introduce foothold, end-effector, object, or force-interaction targets [64–67]. Reference-motion rewards provide a stronger behavioural prior, but introduce demonstration data and are therefore discussed as tracking-based RL in Sec. 4.3.

The humanoid-specific challenge is balancing task progress against physical viability: aggressive task optimisation can destabilise contacts or exceed actuation limits, whereas excessive regularisation can suppress agile behaviour. Recent systems therefore increasingly structure competing objectives rather than simply adding more weighted terms. Examples include separating dense locomotion and sparse foothold value learning, progressively introducing interaction forces through curricula, or decomposing locomotion and end-effector stabilisation across specialised objectives [65–67]. Nevertheless, reward and termination terms remain soft design choices whose behaviour depends on their weights and training coverage, motivating the motion priors and hybrid RL-IL formulations discussed in Sec. 4.3.

4.1.3 Policy Observation and Action Interfaces

A deployable humanoid policy is defined not only by its learning algorithm, but also by what it senses and what it directly controls. Most low-level policies rely on compact proprioception data, including joint and base states, projected gravity, and previous actions, together with task or motion com-

mands. This interface supports reliable high-rate feedback, but remains partially observable: terrain geometry, contact conditions, disturbances, and latent dynamics may not be identifiable from a single observation. Perceptive policies therefore incorporate external sensing such as depth or terrain features, while history-conditioned estimation and reconstruction infer task-relevant state from recent sensorimotor context [68–71].

On the action side, joint position or offset targets tracked by low-level PD control remain the dominant interface for learned humanoid WBC. Compared with direct torque commands, this design places local stiffness and damping inside the servo loop, constrains exploration to a more hardware-compatible control space, and matches common humanoid actuator stacks [54, 59, 60]. Some systems additionally adapt gains or impedance parameters, but the central pattern remains a learned policy that commands joint-level targets rather than raw actuator torques.

More structured action interfaces are useful when direct whole-body control is unnecessarily difficult. Learned policies may instead output footsteps, task-space targets, reduced-order or MPC parameters, or residual corrections around a nominal controller [72–74]. Such interfaces reduce the effective search space and inject useful control structure, but delegate part of physical execution to an external controller; they therefore lie near the boundary between pure learned WBC and the hybrid regimes discussed later.

4.1.4 Policy Architectures for Memory and Behavioural Scaling

Humanoid policy architectures are shaped primarily by two demands: inferring hidden state and scaling to increasingly diverse whole-body behaviours. Compact MLPs remain the practical default for high-frequency low-level control due to their low inference latency and their compatibility with end-to-end RL training [54, 59, 75]. However, it’s hard for them to infer contacts, disturbances, payloads, or latent dynamics from a single observation. Therefore, some works explored adopting history encoders, recurrent policies, and causal Transformers to use recent sensorimotor context for hidden-state inference and longer-horizon adaptation [76, 77]. Such temporal architectures can still be trained end-to-end with RL, although this becomes increasingly challenging as model capacity and behavioural coverage grow.

A second pressure is behavioural interference as one controller expands across diverse motions and contact regimes. Mixture-of-experts (MoE) architectures mitigate this interference by allocating capacity across different regions of the motion manifold; GMT, for example, employs motion-specialised experts within a unified whole-body tracker [78]. A related strategy moves specialisation outside the network, training narrower motion- or embodiment-specific experts before consolidating them into a

higher-capacity generalist [79, 80]. Humanoid-GPT pushes this route further by training PPO experts over motion clusters and distilling them into a causal Transformer, separating physics-grounded RL from large-model consolidation [56]. Architecture design therefore increasingly concerns not only network structure, but also efficient training recipes as model capacity and behavioural coverage scale. The corresponding distillation mechanisms are discussed further in Sec. 4.3.

4.1.5 Curriculum Pipelines and Sim-to-Real Transfer

Training a deployable humanoid policy involves a sequence of coupled bottlenecks: making contact-rich behaviours learnable, exploiting simulation-only information without changing the deployment interface, broadening robustness, and closing the remaining sim-to-real gap. Although these ingredients are often interleaved in practice, they define a useful progression from skill acquisition to hardware deployment.

The first challenge is exploration. Random policies rarely discover stable multi-contact sequences or coordinated loco-manipulation, so curricula progressively expand command, terrain, contact, or interaction difficulty after prerequisite motor skills emerge. State initialisation, staged resets, and contact-stage decomposition further expose otherwise rare but useful states [81, 82]. For coupled loco-manipulation, body-wise decomposition can reduce optimisation interference by separating the robustness-oriented lower-body locomotion objective from the precision- or interaction-oriented upper-body objective [66, 67]. Such decomposition is not a necessary endpoint, however; unified controllers such as ULC recover tighter whole-body coordination through progressive skill acquisition and robustness training [83].

Simulation can further ease optimisation by exploiting information that is available during training but unavailable at deployment. In asymmetric actor-critic learning, the critic may access exact states, contacts, terrain, or dynamics parameters while the actor remains restricted to deployable sensing. A stronger variant first trains a privileged policy and then transfers its behaviour to a deployable perceptual policy [84]; such teacher-student distillation is discussed further in Sec. 4.3.

Once a viable policy is learned, sim-to-real robustness is improved by combining calibration with randomisation: system identification and real-to-sim alignment reduce the nominal mismatch, while domain randomisation over physics, contacts, actuation, sensing, and latency broadens the deployment envelope. Visual policies additionally randomise camera parameters, appearance, image artifacts, and perception delays [84].

Even after robust simulation training, residual sim-to-real mismatch can still emerge on hardware.

While zero-shot transfer remains desirable, limited real-world interaction can be used to correct this mismatch without retraining the policy from scratch. ASAP, for example, learns a residual action model from real humanoid rollouts, feeds the estimated discrepancy back into simulation, and then fine-tunes the policy for redeployment [85]. Together, these practices form a practical progression from simulation-based skill acquisition to robust deployment and targeted real-world adaptation.

4.2 Imitation Learning

4.2.1 Pure Human Motion References

Pure human motion references describe pure human body trajectories without explicit scene, object, or task context. They are commonly obtained from motion-capture (MoCap) datasets or reconstructed from human videos [77, 86–88]. Their scale and diversity make them a major source for acquiring expressive and agile whole-body skills, and they are ideal to be increasingly aggregated into large motion corpora [55, 56].

Before policy learning, raw human motion must be converted into robot-compatible references. They are usually first reconstructed into structured three-dimensional body motion, such as parametric body models, skeletal trajectories, or keypoints. The resulting sequences are then cleaned to suppress noise and temporal discontinuities, and to correct artifacts. The subsequent retargeting stage bridges differences in body proportions, joint topology, and degrees of freedom through inverse kinematics, contact-aware whole-body optimisation, or learned morphology-aware mappings [89, 90]. The resulting supervision typically contains reference joint poses and velocities, root motion, end-effector trajectories, and optional contact labels, rather than the original actions that produced them.

Though these references are inexpensive to scale and can cover motions that would be difficult to collect directly on hardware, they only offer motion style and do not guarantee dynamic feasibility under real execution context, and are therefore not directly suitable for behavioural cloning. Instead, they are usually grounded through RL under balance, contact, and actuation constraints, where they mainly serve as trajectory-level tracking targets, distribution-level motion priors, or extracted as latent motor representations [52, 53, 91]. We will further discuss this in Sec. 4.3.

4.2.2 Context-Rich Human Interaction Data

Context-rich demonstrations preserve not only body motion, but also the scene, objects, and visual observations that make the motion task-relevant. They can be mined from RGB or RGB-D interaction

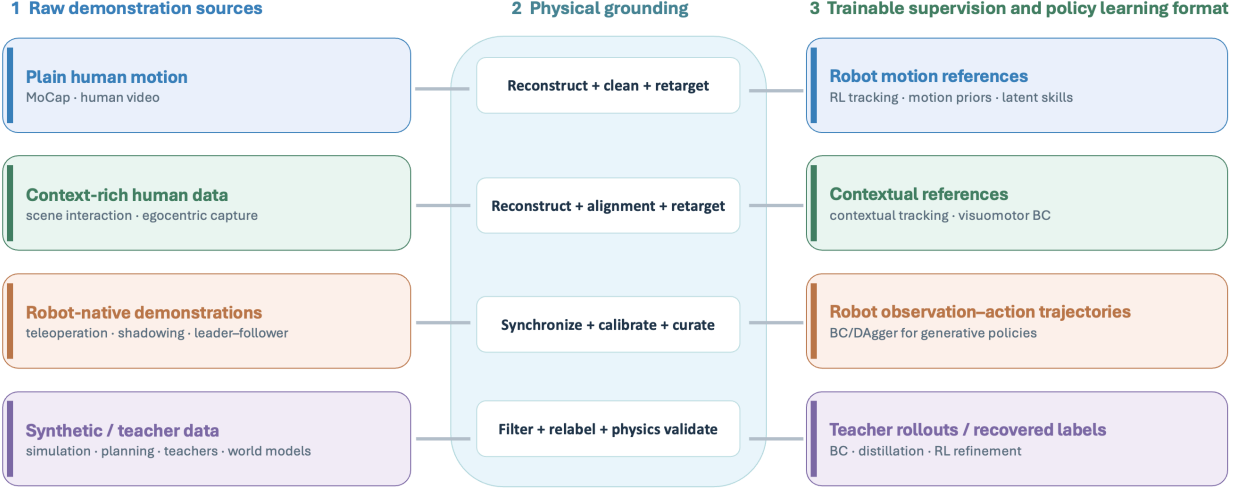


Figure 6: Demonstration sources and their pathways to trainable supervision in humanoid whole-body control. The categories indicate dominant pathways and are not mutually exclusive.

videos, or collected through portable robot-free interfaces that record egocentric vision together with human keypoints, hand trajectories, or gripper commands [92–95]. Compared with pure motion references, these data are especially valuable for locomanipulation and scene interaction, where success depends on how the body moves relative to the surrounding environment.

Their central difficulty is context alignment rather than motion retargeting alone. Human and robot data differ in morphology, camera viewpoint, reachable workspace, object frames, and contact geometry. Transfer must therefore preserve object-relative poses, scene coordinates, visual correspondence, and contact timing while mapping human motion into a robot-feasible reference or action space [93, 94, 96]. A retargeted motion may remain visually plausible yet fail the task if these interaction relations are lost.

After alignment, context-rich demonstrations can remain as reference-level signals containing recovered human, object, and scene motions, and are grounded similarly through RL tracking or other physics-aware controllers [92, 93]. Alternatively, alignment may convert human observations and motions into robot-compatible high-level actions, such as end-effector targets, future keypoints, or locomotion commands, enabling behavioural cloning or VLA co-training while a reusable low-level controller handles balance and whole-body execution [94, 95]. Context-rich data therefore do not eliminate the need for physics grounding; rather, they extend human motion supervision from generic motor priors toward task-conditioned visuomotor learning.

4.2.3 Robot-Native Demonstrations

Robot-native demonstrations are trajectories executed by the target robot, and recorded directly in the robot’s observation and control spaces. Whole-body teleoperation is a prominent collection route

for humanoids, using motion capture, wearable trackers, VR/MR devices, or leader–follower interfaces to map human intent onto robot motion [77, 97, 98]. Unlike human motion references, these demonstrations pass through the robot’s kinematics, hardware limits, and stabilising controllers during collection, allowing visual observations, proprioceptive states, and control commands to be recorded together.

The main processing challenge is temporal and control-stack alignment. Sensor and command streams with different rates and delays must be synchronized; cameras, tracking devices, and robot coordinate frames must be calibrated; and failed episodes, communication interruptions, actuator saturation, and manual recovery segments must be curated. Particular care is required to reconcile commanded and executed actions. An operator may specify end effector targets, base commands, or whole-body poses, which are subsequently filtered and transformed by inverse kinematics or low-level controllers. Such commands provide valid supervision only when they match the action interface of the policy being trained; otherwise, the executed trajectory remains a robot motion reference rather than a complete observation–action demonstration.

When this alignment is preserved, robot-native data provide the shortest path to standard imitation learning because both observations and actions are expressed in the target embodiment and control space. The resulting sequences can support behavioural cloning, interactive correction through DAGger, and sequence or generative policies that predict action or action chunks. They may also initialise latent skill models or demonstration-augmented RL, as discussed in Sec. 4.3. Their main limitation is scalability: collection requires robot access, reliable low-level control, operator effort, and careful safety management, while the resulting data inherit the capabilities and biases of the teleoperation controller.

4.2.4 Synthetic and Teacher-Generated Demonstrations

Unlike the preceding pathways where data generation requires human in the loop, synthetic and teacher-generated demonstrations automate much of the data-production process. Simulation, trajectory optimisation, model-based planners, and learned teachers can generate trajectories repeatedly and in parallel while varying initial states, task layouts, disturbances, and failure conditions to cover rare states and recovery behaviours [99–101]. In practice, they often expand a small set of human demonstrations rather than eliminate human input entirely [102, 103].

These generators provide two main forms of supervision. Reference-level synthesis produces motions, task-space trajectories, contact sequences, or synthetic videos that specify desired behaviour without necessarily providing target-robot actions. Such data still require retargeting, action recovery, or physics grounding. Generative world models offer an emerging example: predicted visual trajectories can be converted into pseudo-actions through inverse dynamics or latent-action inference, although this route has been explored mainly in manipulation and remains less established for balance-critical humanoid control [104, 105]. By contrast, planners, controllers, or learned teachers executed within the robot model can produce synchronized observation–action rollouts suitable for behavioural cloning, sequence-policy learning, or distillation [99, 103]. Reinforcement-learned teachers are discussed further in Sec. 4.3.

Generated data must still be filtered for success and feasibility, aligned with the student’s observation and action spaces, and validated against simulator mismatch [100, 103]. Their scalability is therefore limited by simulator fidelity, teacher competence, and generator coverage. Human-derived and robot-recorded demonstrations remain important anchors for task intent, natural contact strategies, and real perceptual context, while algorithmic generation primarily expands their state, scene, and environment coverage.

4.2.5 Cross-Cutting Policy Learning

Once demonstrations have been converted into policy-aligned observation–action supervision, behavioural cloning provides the most direct learning objective. Its main limitation is covariate shift: small closed-loop errors can move the robot beyond the demonstrated state distribution and compound over time. Interactive aggregation methods such as DAgger mitigate this problem by querying an expert on learner-visited states [106]. For humanoids, however, repeated intervention can be costly or unsafe, so broader offline datasets, synthetic teacher rollouts, and subsequent RL refinement are often used to im-

prove state coverage.

The temporal representation of actions is particularly important for high-dimensional whole-body behaviour. Rather than predicting isolated actions, action-chunking policies generate short sequences of joint targets, task-space commands, or future key-points, which are executed with receding-horizon re-planning or temporal aggregation to preserve coordinated intent and reduce jitter [77, 107–109].

Standard regression is effective when demonstrations are locally consistent, but may average across multiple valid behaviours. Diffusion and flow-based policies instead model conditional distributions over action sequences, supporting multimodal behaviour generation [58, 110]. Such generative policies are well established in manipulation and are beginning to appear in whole-body humanoid interaction and loco-manipulation, although they remain less mature than conventional chunked imitation policies [111, 112].

4.3 Demonstration–RL Integration

4.3.1 Reference-Tracking RL

When demonstrations enter RL as explicit trajectory targets, the policy is optimised to follow a time-indexed reference while task and physical rewards enforce balance, contact feasibility, and actuation constraints [52]. This is the most direct form of demonstration guidance: the reference specifies the desired whole-body behaviour, while RL learns the feedback corrections required to execute it under humanoid dynamics. In data-processing pipeline discussed in Sec. 4.2, we’ve argued that tracking-based RL is useful for grounding reference motion intents into real-world feasible robust controllers. In this section, the key question is how broadly a single feedback controller can track such references.

Reference tracking can reproduce new skills from limited or even single demonstrations, making it particularly effective in low-data settings [113, 114]. Beyond such skill-specific adaptation, recent work has progressed toward increasingly general motion trackers that cover broad manifolds of locomotion, expressive motion, and highly dynamic whole-body skills [75, 115–117]. Such trackers also provide a natural interface for real-time whole body teleoperation, where live human motion is continuously translated into executable references [77, 97, 118]. The same principle is extending to interaction-aware tracking, where successful execution must preserve object motion, contact timing, and coupled human–object dynamics rather than body pose alone [119, 120]. References themselves may also be generated online, where diffusion or autoregressive models propose task- or terrain-conditioned motion trajectories that are subsequently grounded by a low-level RL tracker [121–124]. Most importantly, dense reference-based rewards make motion tracking a favourable substrate

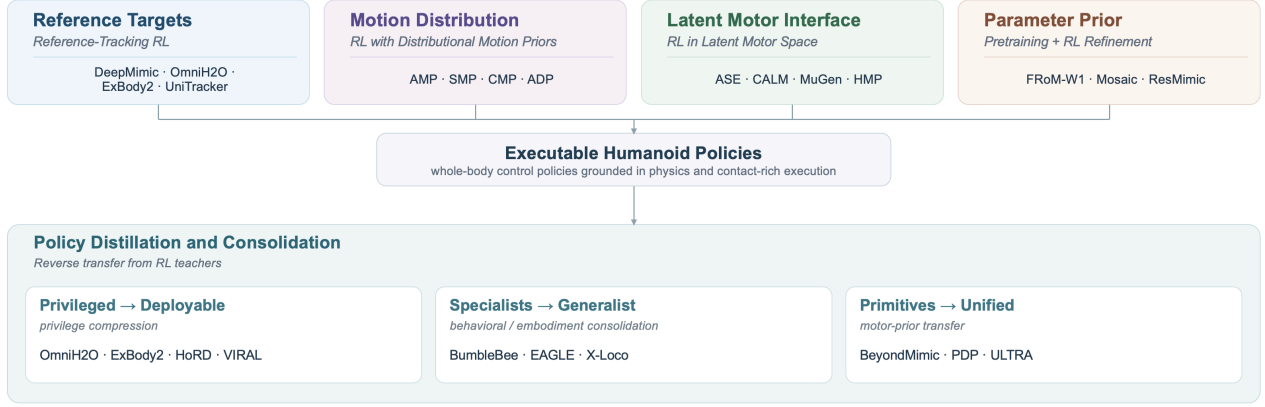


Figure 7: Demonstration-RL integration in humanoid whole-body control. Demonstrations can guide RL through explicit reference targets, distributional motion priors, latent motor interfaces, or pretrained policy parameters, yielding executable whole-body policies. Policy distillation reverses this information flow, using learned controllers as supervision for deployable students, generalist controllers, or alternative unified policy representations. Categories denote dominant roles and are not mutually exclusive.

for scaling RL task coverage and network capacity. At the frontier, SONIC and Humanoid-GPT treat tracking as a scalable pretraining task, expanding motion data and model capacity to improve coverage and zero-shot generalisation [55, 56]. This trajectory toward reusable motor foundation models is revisited in Sec. 4.4.

4.3.2 Distributional Motion Priors

When demonstrations enter RL as a distributional prior rather than an explicit trajectory target, they constrain how the humanoid should move without prescribing a particular reference sequence. Adversarial Motion Priors (AMP) provide the canonical formulation: a discriminator distinguishes policy-induced transitions from reference motion, and its score becomes a motion-style reward combined with the task objective [53]. Compared with reference tracking, this removes the need for exact phase alignment or clip-wise tracking and allows the task reward to select behaviours within the demonstrated motion distribution. The trade-off is weaker control over timing, contacts, and task-specific motion details, making the quality and relevance of the learned prior increasingly important.

Recent work broadens motion priors beyond task-agnostic adversarial discrimination. Score-matching methods learn smooth motion-density or energy models, and SMP further turns such a prior into a task-independent reward that can be frozen and reused across downstream controllers [125, 126]. This reusability, however, creates a new problem: motions favoured by a global prior may be irrelevant or even conflict with the current task. CMP addresses this by weighting adversarial and score-based guidance according to context-dependent motion relevance [127]. A parallel direction introduces dynamics-aware priors over quantities such as centroidal motion and contacts [128]. Together, these

developments move distributional guidance toward more reusable, task-aware, and physically grounded motion priors.

4.3.3 Latent Motor Interfaces

When demonstrations define the action space rather than the reward, their motor competence can be compressed into a latent interface. A low-level controller maps a compact code z to physically feasible whole-body actions, while a downstream RL policy selects or modulates z instead of directly controlling all joints. This reduces the search space for high-dimensional, contact-coupled humanoid control and constrains exploration to a learned motor manifold. Common representations include continuous latent spaces learned through variational models such as VAEs/CVAEs or adversarial skill embeddings, as well as discrete skill spaces constructed through VQ-VAE or residual vector quantisation [91, 98, 129, 130].

Recent work increasingly grounds these representations in executable motor competence rather than motion reconstruction alone. Tracking policies or other physics-grounded experts can first acquire robust whole-body skills and then be distilled into a latent encoder, codebook, and action decoder. HMP, for example, distills a motion-tracking expert into a frozen RVQ motor interface, after which downstream RL policies solve new locomotion tasks by selecting discrete codes rather than joint actions [131]. Such interfaces improve skill reuse and downstream exploration, but restrict control to the learned motor manifold and may hide contact-level details needed by new tasks. Their use as interfaces for language and multimodal models is discussed further in Sec. 4.4.

4.3.4 Pretraining and RL Refinement

When demonstration supervision is absorbed into a controller before downstream optimisation, it guides RL through the policy parameters themselves. A pretrained tracker or motion controller already encodes coordinated whole-body behaviour, allowing subsequent RL to focus on task success, interaction, or deployment-specific constraints rather than rediscovering basic motor competence. This creates a central trade-off between adaptation and prior preservation. Full-policy fine-tuning provides the greatest flexibility; FRoM-W1, for example, pretrains a general whole-body motion controller and then applies RL fine-tuning to improve tracking accuracy and physical execution [132]. However, unrestricted updates may narrow the learned behavioural prior or degrade motions outside the refinement distribution.

A more conservative strategy preserves the pretrained controller and learns only task- or domain-specific corrections. Residual policies can refine nominal actions for object interaction, contact precision, or deployment mismatch while retaining the broad motor competence of the base policy. ResMimic keeps a general motion tracker as a task-agnostic backbone and learns a residual policy for precise loco-manipulation [133], while MOSAIC uses rapid residual adaptation to specialise a general tracker to interface and dynamics discrepancies without sacrificing its generality [134]. The key design choice is therefore how much of the demonstration-derived prior should remain fixed as RL adapts to the target setting.

4.3.5 Policy Distillation and Consolidation

Distillation reverses the preceding direction of demonstration-guided RL. Rather than using demonstrations to shape RL, an executable teacher which is commonly trained by RL now provides action demonstration for a student policy. In humanoid WBC, its most established role is privilege compression. Teachers trained with privileged simulator states, future references, or environment information can acquire robust whole-body skills, while students imitate their behaviour using only deployable sensing, preserving the motor competence without the privileged deployment interface [75, 76, 84, 97]. This provides a practical bridge from simulation-optimised controllers to deployable humanoid policies.

Distillation increasingly also serves behavioural consolidation. One pattern aggregates multiple specialists into a generalist, combining expertise across motion families, embodiments, or control regimes [79, 80, 135]. A distinct pattern transfers physically grounded competence from RL trackers or motion primitives into a different policy representation, where diffusion or other multimodal students imitate executable rollouts and can subsequently

compose or condition the acquired behaviours for broader tasks [136, 137]. Distillation therefore closes the demonstration-RL loop in the reverse direction, turning learned controllers themselves into supervision for more deployable, general, or reusable policies.

4.4 Foundation Models for Humanoid Whole-Body Control

Foundation models entered robot learning largely through vision-language-action (VLA) models, which leverage large-scale multimodal data to learn direct mappings from visual and language inputs to robot actions through end-to-end supervised imitation learning [57, 58, 138]. While this direct recipe is well established for the manipulation regime, humanoid whole-body control faces harder challenges of high-dimensional actions, balance-critical and contact-rich environments, and locomotion-manipulation constraints at high feedback rates [139, 140]. Humanoid systems therefore often acquire multimodal understanding, predictive world knowledge, and robust motor competence separately before coupling them through hierarchical or unified control architectures [141–143]. We organise this section accordingly around two questions: which reusable modular capabilities are learned, and how are they coupled to physics-grounded whole-body execution.

4.4.1 Reusable Foundation Capabilities

Multimodal understanding provides reusable semantic and perceptual knowledge about what the robot should do. This capability largely inherits from manipulation-side VLM/VLA pretraining, where large-scale vision, language, and robot data support object understanding, instruction following, task reasoning, and visuomotor generalisation [57, 58, 138]. For humanoids, the same knowledge can specify task intent, relevant objects, and desired interactions, but does not by itself resolve how these intentions should be executed under balance and contact constraints. Multimodal pretrained foundation models are therefore adopted primarily as a reusable task-understanding prior that provides guidance for low-level action generators.

Predictive world modelling adds knowledge of how the observed world is likely to evolve under action. Rather than mapping only the current observation to an action, world models learn action-conditioned future observations or compact future representations, enabling a policy to reason about possible consequences before or while acting. Foundation-scale world modelling has so far developed more strongly in manipulation and general robotics, where large-scale video pretraining supports simulation, evaluation, and planning [144–146]. Its use in humanoid WBC remains comparatively emerging, but recent world-action models begin to

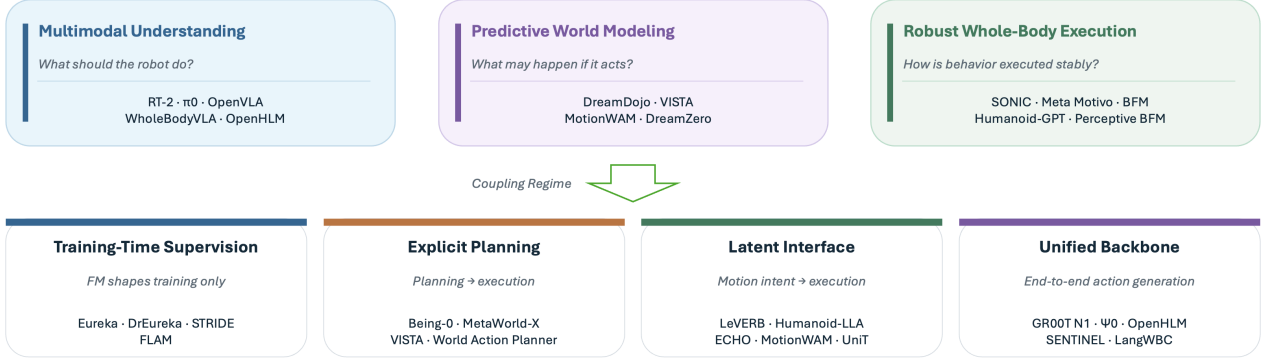


Figure 8: Overview of humanoid WBC foundation models.

couple visual foresight with whole-body motion generation for loco-manipulation [112, 147]. Predictive modelling is therefore a complementary rather than necessary capability: it extends reactive control with foresight about future interaction outcomes.

Robust whole-body execution concerns how physically feasible behaviours should be generated. Humanoid control has long acquired this competence through tracking-based RL discussed in Sec. 4.3.1. The foundation-model shift is to scale such experience from task-specific controllers toward broad motor priors reusable across motions, commands, and downstream behaviours. Recent behavioural foundation models accordingly scale motion diversity, physics-grounded rollouts, and policy capacity to obtain increasingly general whole-body controllers [55, 56, 148–150]. Unlike multimodal foundations, their primary transferable asset is executable motor competence rather than semantic and visual task knowledge. In modular systems, such motor foundations naturally serve as robust executors beneath higher-level multimodal models that interpret the scene, select task intent, and generate motion-level commands. How such hierarchical capabilities are coupled forms the focus of the next section.

4.4.2 Coupling Foundation Capabilities in Humanoid Control

Having acquired reusable semantic, predictive, and motor capabilities, the remaining design question is how deeply they should be coupled within the humanoid control stack. Existing systems span a continuum from foundation models used only during training to policies that internalise multimodal reasoning and action generation within a unified deployment architecture.

Training-time supervision represents the weakest form of coupling: foundation knowledge shapes policy learning but does not remain in the deployed control loop. Language and multimodal models can automate reward design, task specification, and training feedback [151–153], while pretrained motion models can provide physically meaningful stabilisation or motion-quality supervision for RL [154].

Predictive world models can play a similar role by generating imagined experience or evaluating candidate behaviours. This regime preserves a conventional high-rate controller at deployment while exploiting foundation-scale knowledge to improve how it is trained.

Explicit planning and skill interfaces retain foundation models at deployment but separate reasoning from execution through interpretable sub-goals, skills, or task-level commands. A high-level model can decompose instructions and select among reusable motor experts, leaving balance-critical feedback control to specialised policies [155, 156]. Predictive world models naturally extend this architecture by evaluating or refining candidate plans through imagined futures before execution [145]. Such hierarchical designs preserve a clear separation between slower semantic or predictive reasoning and fast physical control.

Motion and latent interfaces move the coupling boundary closer to the motor layer by replacing discrete skills with trajectories, motion tokens, or continuous latent commands. These representations provide a compact bridge between semantic intent and high-rate whole-body control, allowing the upstream model to specify richer behaviours while a dedicated executor preserves balance and contact robustness [139, 140, 157]. Predictive world models can be coupled at the same level: unified physical latents can encode anticipated interaction outcomes [158], while MotionWAM uses video-world-model features to generate whole-body motion tokens that are subsequently realised by a robust motor controller [112]. This interface is particularly natural for humanoids because it reconciles flexible task reasoning with the substantially faster feedback rates required for physical execution.

Unified direct-action coupling removes an externally exposed planning or motion interface and allows multimodal representations to condition robot actions directly. Importantly, this does not imply a monolithic network or end-to-end training from scratch: internal vision-language and action modules may still be specialised and jointly trained, pre-trained in stages, or consolidated through distillation

Table 6: The four regimes of learning-based whole-body control along the data-diversity axis κ_d .

Regime	Data anchor	Supervision	Sample efficiency	Generalisation	Sim-to-real burden	Representative families
Pure RL	reward-sampled rollouts	hand-designed reward	low (from scratch)	task-bounded	high (domain rand.)	terrain / parkour locomotion
Pure IL	expert demonstrations	action supervision	high (offline)	demonstration-bounded	moderate (covariate shift)	BC, DAgger, diffusion policy
RL + IL	demonstrations + reward	motion prior + reward	moderate	broad within the prior	moderate	tracking RL, AMP, distillation
Foundation	web-scale multimodal	pretraining + adaptation	amortised	semantic, open-world	highest (contact-rich)	dual-system / token VLA

[141–143, 159, 160]. What distinguishes this regime is that the coupling between task understanding and motor generation is internalised by the deployed policy rather than mediated by a hand-designed skill or latent interface. Manipulation-side world-action models suggest that predictive dynamics may eventually be internalised in the same manner [146], although this direction remains comparatively emerging for humanoid whole-body control.

To sum up, more modular designs are generally easier to optimise and preserve strong task-specific physical robustness, whereas tighter integration offers greater potential for cross-task generalisation but requires more data and harder optimisation. Motion or latent interfaces have therefore emerged as a particularly active middle ground in humanoid control, while direct whole-body VLAs are also advancing rapidly. The field is thus still exploring the optimal boundary between reusable reasoning and robust physical execution.

4.5 Structural Bottlenecks

Three bottlenecks persist across the data-diversity spectrum, and they survive even at the foundation-model end.

The first is *opaque failure*. Because the feasibility constraint g is compiled implicitly into θ , a learned controller offers no inspectable account of why it acts, and a fault surfaces as a fall or a collision rather than as a violated constraint that a designer can read off and repair. More data widens coverage but does not make the policy legible.

The second is the *absence of formal guarantees*. Reward shaping, adversarial priors, and curricula bias a policy toward safe and natural behaviour, yet none of them certifies a stability margin or a hard constraint the way an explicit program does. Safety is encouraged in expectation, not enforced.

The third is the *online-adaptation gap*. The optimisation runs offline, so a deployed policy cannot re-solve its problem when the world leaves the training distribution. Domain randomisation and history-conditioned adaptation broaden the envelope, but the gap reopens whenever a new payload, terrain, or contact mode appears.

These limits mirror the model-based ceiling in reverse. The analytical pillar of Sec. 3 owns an explicit, certifiable, online-replanning solver but cannot generalise from data; the learning pillar owns the generalisation but forfeits that explicit, certifiable, replanning core. Neither end holds both at once. The hybrid methods of Sec. 5 keep the analytical scaffold

exactly where these three bottlenecks bite, and we turn to them next.

5 Hybrid Model–Learning Whole-body Control

5.1 A Topological Taxonomy of Hybrid WBC

Model-based whole-body control (WBC) has matured into a principled framework that combines rigid-body inverse dynamics, weighted quadratic programming, and predictive control to enforce explicit physical constraints and expose interpretable cost terms [8–10]. Its performance is, however, bounded by the fidelity of the model it relies on. On compliant terrain, with uncertain payload, or with rich visual context, the reduced-order assumptions of LIP/ALIP, single-rigid-body, or centroidal models break down, and the controller becomes either conservative or unsafe. Learning-based WBC, primarily reinforcement learning (RL) and motion imitation, has, on the other hand, achieved striking agility and sensor-rich generalisation [1, 11, 12], at the documented price of sample inefficiency, opaque failure modes, the absence of formal guarantees, and a sim-to-real gap that typically widens with task complexity. A growing body of work hybridises the two. “Hybrid” is not a single method: a meaningful taxonomy must distinguish *how* the model-based cluster $\mathcal{M}$ and the learned cluster $\mathcal{L}$ meet. We organise the surveyed works along their *structural relationship* and present them in order of increasing coupling between $\mathcal{M}$ and $\mathcal{L}$. Four mutually exclusive structures (Fig. 9) cover the surveyed literature:

Hierarchical $\mathcal{L}$ and $\mathcal{M}$ operate at different abstraction levels; the interface is a narrow, unidirectional setpoint or reference. Either $\mathcal{L} \rightarrow \mathcal{M}$ (learning plans, model tracks) or $\mathcal{M} \rightarrow \mathcal{L}$ (model abstracts, learning specialises).

Switching $\mathcal{L}$ and $\mathcal{M}$ are independent, complete controllers; at runtime a router selects exactly one. The interface is a discrete activation signal.

Parallel $\mathcal{L}$ and $\mathcal{M}$ emit actions at the same instant; the executed command is a weighted sum or residual composition.

Embedded $\mathcal{M}$ contains $\mathcal{L}$ as an internal module (*e.g.*, a learned cost surrogate inside MPC), or $\mathcal{L}$ contains $\mathcal{M}$ (*e.g.*, differentiable MPC as a

network layer). The two share a computation graph; neither is meaningful in isolation.

The categorical boundary that most often blurs in practice is that of *Embedded* against the other three categories. We adopt a sharp criterion: if removing $\mathcal{L}$ leaves $\mathcal{M}$ mathematically well defined and runnable with default parameters, the interface is Hierarchical, Switching, or Parallel; if removing $\mathcal{L}$ leaves $\mathcal{M}$ mathematically incomplete (because $\mathcal{L}$ supplies the cost function, the dynamics, the Lyapunov function, or a differentiable layer), the interface is Embedded.

5.2 Hierarchical Hybrids

Hierarchical hybrids are the principal model-learning hybrid form, distinguished by *what kind of signal* the high level emits. The three families fall along an abstraction axis: **H1** (plan-then-track) emits a low-dimensional plan, **H2** (decide-then-execute) emits a discrete decision, and **H3** (semantic-then-physical) emits a semantic target. All three share a single architectural prior: *plan in a low-dimensional, semantically meaningful space; act in the full-order physical space*. The further the high-level signal sits from the joint-space action, the more specialised the interface and the heavier the downstream model-based stack must be to bridge the gap.

Two observations bind the three families together. First, the abstraction gap between $\mathcal{L}$ and $\mathcal{M}$ widens monotonically from H1 to H3: contact sets and continuous references live in the same metric space as the controller’s tracking error, discrete skill identifiers live in a quotient of that space, and semantic targets live in a space shaped entirely by the upstream learned encoder. Second, this widening is mirrored by the difficulty of designing the interface itself: a contact set is verifiable by kinematics, a skill identifier by a hand-written guard, but a semantic target requires either a calibration phase or a learned grounding model to be usable by $\mathcal{M}$. The remainder

of this section reviews each sub-type with representative examples and notes the development trajectory within each.

5.2.1 H1: Plan-then-Track

H1 is further split by the content of the plan into three sub-types: a discrete contact set (H1a), a continuous reference trajectory (H1b), or a modulation of the model-based controller’s parameters (H1c).

H1a: Footstep and Contact Planning (6 works). The learned high level emits the location of the next contact, and a model-based controller tracks it via whole-body QP or MPC. Two designs converge on the same principle—a reduced-order template is most useful when it constrains, but does not over-constrain, the policy: Bang et al. [161] residualise an ALIP-based MPC footstep with an RL refinement ($\pi = \pi_0 + \pi_r$), while Lee et al. [162] use LIP-derived footsteps as *partial guidance* that the policy must reach but is otherwise free to deviate from. Dhédin et al. [163] extend the paradigm from single footstep to discrete contact plan, using a diffusion model to distil multi-modal MCTS+NMPC expert solutions for stepping-stone environments; Kim et al. [164] replace the hand-engineered perception pipeline with an RL policy operating on raw elevation maps; Kumagai et al. [165] push the paradigm into multi-contact humanoid planning, with RL acting as a *preference function over candidate contacts* rather than a residual on a single plan. Suliman et al. [166] ablate the template itself, showing that a Raibert-style heuristic combined with RL matches LIP-based designs, hinting at progressive template simplification as RL absorbs more of the implicit dynamics.

H1b: Reference Trajectory and Motion Prior (11 works). The learned high level emits a continuous reference (joint, task, or centroidal) which

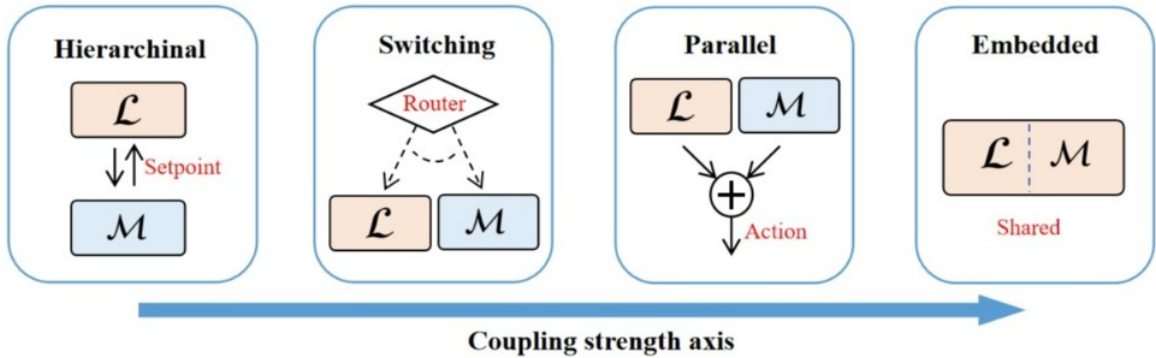


Figure 9: Structural taxonomy of hybrid model-learning whole-body control, ordered left-to-right along coupling strength. (a) *Hierarchical* links $\mathcal{L}$ and $\mathcal{M}$ through a setpoint at different abstraction levels; (b) *Switching* keeps them as complete controllers and selects one via a router; (c) *Parallel* sums their actions at the same instant; (d) *Embedded* fuses them in a shared computation graph.

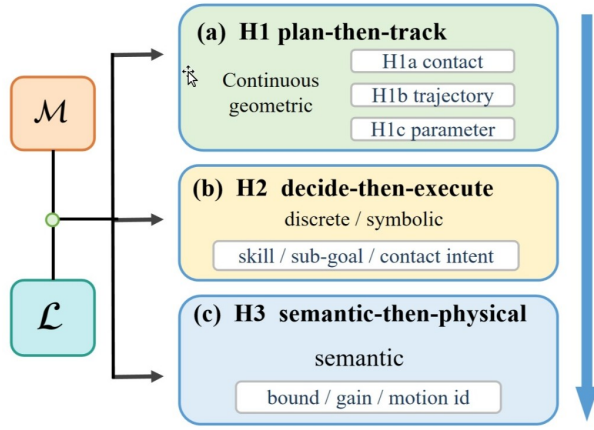


Figure 10: Hierarchical hybrids of Sec. 5.2, ordered top-to-bottom along the abstraction axis. The direction is not fixed: either $\mathcal{L} \rightarrow \mathcal{M}$ (learning plans, model tracks) or $\mathcal{M} \rightarrow \mathcal{L}$ (model abstracts, learning specialises). What is shared is a narrow unidirectional setpoint r between two abstraction levels; the downstream controller remains complete if the upstream module is removed. The three families are ordered by the *type* of r , not by who emits it. (a) H1 carries a continuous or geometric plan (H1a contact, H1b trajectory, H1c parameter). (b) H2 carries a discrete skill, sub-goal, or contact intent. (c) H3 carries a semantic target (bound, gain, or motion identifier) that must be grounded. Interface design grows harder as r leaves the tracking-error metric space.

the low level tracks. Two design philosophies co-exist. The first uses model-based trajectories as a *warm start* for an otherwise standard RL pipeline: the lineage traces back to Levine and Koltun [167], whose Guided Policy Search distils trajectory optimisation (TO) experts into neural policies, and is continued by Bogdanović et al. [168] and Batke et al. [169], who pretrain RL with TO demonstrations to bypass cold-start exploration without committing the deployed policy to track those demonstrations exactly. The second philosophy keeps the model-derived trajectories *in the loop* as an explicit reference to be mimicked, treating trajectory optimisation as a structured prior over feasible motions. Miller et al. [170] (MIMOC) and Fuchioka et al. [171] (OPT-Mimic) imitate libraries of TO-derived trajectories; Liu et al. [172] elevate the prior to whole-body dynamically-feasible trajectories from full-order optimisation; Taouil et al. [173] use a text-conditioned latent diffusion model to generate physically-consistent loco-manipulation references; and Margolis et al. [174] address compliance by training the tracking policy on example motions with low admittance, learning soft rather than stiff regulation. An ablation by Chaikovskaya et al. [175] systematically compares with/without-reference RL training, confirming that model-derived references shorten training time but can ossify the policy if tracked too rigidly. Read across the sub-family, the

trajectory is a steady relaxation of the prior: from hard tracking targets, through example-set mimicking, toward soft, distribution-level guidance from generative models.

H1c: Parameter and Task-Space Modulation (7 works). The learned high level does not produce a plan; it *adjusts the parameters* of an otherwise-complete model-based controller. The parameters may be the task-space command for an inverse-dynamics QP [176, 177], the gait parameters of an MPC [178], the cost weights of an optimal controller [179, 180], or a learned trajectory modulator on top of a template tracker [181, 182]. The defining property of H1c, which separates it from Embedded, is that the model-based controller remains mathematically well defined and runnable with default parameters when $\mathcal{L}$ is removed; training of $\mathcal{L}$ treats $\mathcal{M}$ as a black-box environment. The advantages are practical: training pipelines stay modular, deployment is straightforward, and the prior that the cost / gait structure imposes makes the search space small enough for sample-efficient RL or Bayesian optimisation.

5.2.2 H2: Decide-then-Execute

The learned high level emits a *discrete* or *symbolic* decision—a skill identifier, sub-goal, or contact intent—and a per-decision controller, typically model-based, executes. Two sub-flavours are visible. The first targets gait and foothold selection on top of an optimal whole-body controller: Krishna et al. [183] use oracle guidance to select among hand-engineered locomotion skills, Jenelten et al. [4] (DTC) train an RL policy that adjusts intermediate set-points of a trajectory-optimisation controller for off-road quadruped locomotion, and Gangapurwala et al. [184] (RLOC) cast RL as a foothold-and-gait selector that sits above an optimal whole-body tracker. The second sub-flavour pushes the discrete interface toward task semantics: Kedia et al. [185] train a cost-aware human-motion forecaster that selects the cost function of an MPC for safe human-robot collaboration, and Xie et al. [186] use a hierarchical RL-MPC design in which the high level reasons about *where* and *how* to make contact in dexterous manipulation. Across both sub-flavours, the discrete interface provides a clean abstraction for long-horizon tasks but requires either a pre-existing skill library or careful skill-discovery training; failure modes typically arise at the transition boundary rather than within any individual skill.

5.2.3 H3: Semantic-then-Physical

Foundation-model-guided WBC is the most recent sub-type. A vision-language model (VLM) or large language model interprets the scene and emits a *semantic* signal—a velocity bound, an impedance gain,

a constraint, or a target motion identifier—which a model-based whole-body controller executes. Fernando et al. [187] couple a VLM-RAG (retrieval-augmented generation) module with a whole-body MPC for bimanual mobile manipulation, with the VLM emitting per-task virtual impedance and velocity bounds. Sleiman et al. [188] use text- and video-conditioned imitation to produce zero-shot embodied skills tracked by an RL whole-body policy. He et al. [189] demonstrate an end-effector-centric teleoperation framework for aerial manipulation where the human (effectively an external semantic source) commands the end-effector pose and a whole-body controller resolves redundancy. The common thread is the use of a high-capacity perception-language module as a non-differentiable, task-specifying oracle for an otherwise classical controller. Two open architectural questions are visible already at this small sample size: whether the foundation model should emit *parameters* of the controller (as in Fernando et al. [187], where impedance and velocity bounds are sampled), *references* for the controller (as in Sleiman et al. [188], where motion identifiers seed an RL tracker), or *constraints* on the controller (a path not yet taken in the surveyed corpus); and, orthogonally, whether the semantic-to-physical translation should happen at the policy boundary or be pulled inside a differentiable controller, in which case the design migrates toward the embedded category. We expect H3 to be the locus of the most visible methodological change in the next 24 months.

5.3 Switching

Switching hybrids replace continuous interfaces with a discrete activation signal: $\mathcal{L}$ and $\mathcal{M}$ (or two $\mathcal{L}$ ’s, or two $\mathcal{M}$ ’s) are complete controllers, and a router toggles which one is in command. Baek et al. [190] (BAT) train an online policy switcher that trades off agility and stability across long-horizon humanoid tasks, learning when to commit to a high-energy skill versus when to default to a stabilising controller. He et al. [191] (Agile But Safe) compose an agile RL policy with a recovery RL policy under a learned safety advisor, achieving high-speed collision-free locomotion through reactive policy switching. Yu et al. [192] learn the switching criteria themselves over a heterogeneous skill set for quadruped locomotion. Switching is structurally simpler than parallel or embedded fusion but relies on the existence of multiple competent sub-controllers and on a reliable router; its safety properties are inherited from those of the active sub-controller at each instant, plus the discrete decision of the router. The principal design degrees of freedom are: the *dwell time* the router must enforce to avoid chattering between sub-controllers, the granularity of the switching signal (per-step versus per-skill), and whether the router itself is learned end-to-end or trained as a separate classifier on a labelled corpus of safe and unsafe states. Conceptually,

switching can be read as a hard-assignment limit of parallel fusion, in which the weight collapses to $\{0, 1\}$; this also suggests a natural relaxation path—soft-switching with a learned, possibly stochastic gating—that is so far under-explored in WBC.

5.4 Parallel and Residual

Parallel hybrids let $\mathcal{M}$ and $\mathcal{L}$ contribute simultaneously to the executed action. The most common pattern is *residual*: $\mathcal{M}$ produces a nominal action, $\mathcal{L}$ produces a correction, and the two are summed. Jeon et al. [3] blend a GPU-parallelised MPC with a reinforcement-learned residual that compensates for the MPC’s modelling errors. He et al. [193] (HOVER) train a versatile neural whole-body controller in which multiple task-specific RL heads operate in parallel atop a shared nominal model. Schperberg et al. [194] compose a model-based loco-manipulation controller with a learned safety/perception branch, with the two clusters serving complementary stability and adaptation roles. Li et al. [195] train RL to track arbitrary trajectories in parallel with the model-based tracking controller, and Bi et al. [196] (ALORE) introduce a residual learned adjustment for autonomous large-object rearrangement with a legged manipulator. The principal advantage of parallel hybrids is that the safety properties of $\mathcal{M}$ are preserved as long as the residual $\mathcal{L}$ is bounded; the principal challenge is the action-level coordination between the two branches, since each must remain valid without knowledge of the other’s instantaneous choice. A second axis distinguishes *additive* residuals—in which $\mathcal{L}$ contributes a separately bounded correction Δa trained against the deployed $\mathcal{M}$ in the loop—from *differentiable* residuals, in which gradients flow through $\mathcal{M}$ during training and the residual is co-optimised with the model’s parameters; only the former is in scope here, with the latter belonging to the embedded category. In practice the additive formulation is also the easiest hybrid pattern to retrofit onto a mature model-based stack, which explains its growing popularity for production-oriented humanoid loco-manipulation.

5.5 Embedded

Embedded hybrids fuse $\mathcal{M}$ and $\mathcal{L}$ at the level of the mathematical formulation: $\mathcal{L}$ supplies the cost, the value, the dynamics, the Lyapunov function, or a differentiable layer that $\mathcal{M}$ uses internally; alternatively, $\mathcal{M}$ appears as a layer of a network that is trained end-to-end. Five sub-types are visible in the surveyed literature.

Learn-the-cost / learn-the-value. Romero et al. [2] embed an RL critic value as the terminal cost of a differentiable MPC, jointly training the policy and the critic with gradients propagated through the optimisation. Li et al. [197] learn a control-Lyapunov

function as a neural network and use its prescribed decrease condition as a constituent of the RL reward, so that the CLF and the policy co-evolve.

Learn-the-model / learn-the-safety. Tay et al. [198] couple a centroidal dynamics simulator inside the RL forward pass for humanoid motion mimicking, letting the policy reason about centroidal momentum at training time. Ahn et al. [199] use a learned dynamic balancing model to shape RL training. Yang et al. [200] learn the dynamics that appear inside a control-barrier function, with constraint satisfaction defined in expectation over the learned model.

Differentiable optimisation as a network layer. Xiao et al. [201] treat MPC as a layer in a real-time-attention navigation network, training end-to-end on demonstration data.

Physics-guided generative models. Cho et al. [202] constrain a rectified-flow text-to-motion generator with physics gradients so that sampled motions are kinematically and dynamically feasible at the controller level. Araújo et al. [203] use differentiable physics in the loss of a motion-retargeting network so that retargeted motions remain trackable on the target humanoid.

MBC inside RL training (as expert, teacher, or critic). Peng et al. [204] use an MPC expert to teach a generalist–specialist distillation pipeline; Wang et al. [205] pretrain a humanoid WBC policy with model-based reference data and a fast model-based adaptation module; Zhong et al. [206] (BAS) bridge adaptivity and safety with a model-based safety branch inside the RL training loop; Zhang et al. [207] use a foundation-model-derived body-stabilisation prior during pretraining. Across this sub-type the model is present at training time but the deployed policy is single-network; the boundary with H1b (motion prior) is determined by whether gradients flow through the model-based cluster (embedded) or only its outputs are imitated (H1b).

Read across the five sub-types, embedded hybrids share a common methodological signature: the design problem ceases to be “how do $\mathcal{L}$ and $\mathcal{M}$ exchange signals?” and becomes “how do we co-design and co-train them?”. This explains both their expressiveness—the safety, dynamics or cost structure of $\mathcal{M}$ can be enforced *by construction* rather than hoped for—and their engineering cost: stable end-to-end training through a constrained optimisation, a CLF, or a learned dynamics model typically requires custom gradients, careful conditioning of the forward solve, and dedicated debugging tools, all of which are still research artefacts rather than commodity infrastructure.

5.6 Comparative Discussion and Outlook

The distribution of the corpus across the four categories is highly skewed: hierarchical hybrids dom-

inate with 34 of 55 works (62%), followed by embedded (14, 25%), parallel (4, 7%) and switching (3, 5%). The temporal trend is even more pronounced (Fig. 11): 38 of the 55 surveyed works (69%) are from 2024–2026, and embedded hybrids concentrate in the same window, with nine of fourteen appearing from 2024 onward. The earlier instances, beginning with Ahn et al. [199], predate the differentiable-MPC and GPU-parallel sampling-MPC tooling that catalysed the rest, with the H3 (semantic) sub-family appearing only in the last twelve months as VLM and text-to-motion models matured into a viable upstream interface for whole-body controllers.

Table 7 summarises the trade-offs across the four super-classes. Three observations emerge.

First, *no single category dominates*: Hierarchical wins on modularity and sample efficiency, Embedded wins on enforceable safety and structural guarantees, Parallel is the lowest-friction upgrade path for a mature model-based stack, and Switching is the natural choice when the task itself decomposes into qualitatively different regimes. Second, *coupling strength correlates with engineering complexity*: as one moves from Hierarchical to Embedded, the training pipeline becomes denser (differentiable MPC, implicit function theorems, joint training of dynamics and policy), increasing both potential and brittleness. Third, the *H1c–Embedded boundary* discussed in Sec. 5.1 is the most consequential in practice: choosing to expose the model only as a black-box parameter sink (H1c) versus to differentiate through it (Embedded) often determines whether a hybrid stack can be assembled by a small team in months or requires a multi-year research effort. Fourth, an aggregate observation about *evaluation*: across the four categories, hardware experiments are still predominantly run on a single platform per paper, and quantitative comparisons across hybrids are rare; even within a single category, the choice of robot, simulator and task overwhelmingly dominates the apparent performance gap, so claims of “best” hybrid architecture should be read with category-level context rather than as platform-independent rankings.

Five directions stand out for future work on hybrid WBC. (i) *Bidirectional information flow*. Almost all surveyed hierarchies are downward; how a model-based low level can systematically inform the learned high level (about constraint activation, force margins, or contact transitions) remains under-explored. (ii) *Theoretical guarantees for hybrid stacks*. Stability, safety, and recovery proofs typically address either $\mathcal{M}$ alone or $\mathcal{L}$ alone; the interface itself introduces unmodelled effects (residual dynamics, switching dwell time, learned-cost smoothness) whose analysis is still ad hoc. (iii) *Standardised benchmarks for hybrid WBC*. The surveyed literature is fragmented across robot platforms, simulators, and tasks; comparative claims are difficult to verify. (iv) *Integration of foundation models*. The H3 sub-class is nascent; how VLMs/LLMs can serve as principled modules of

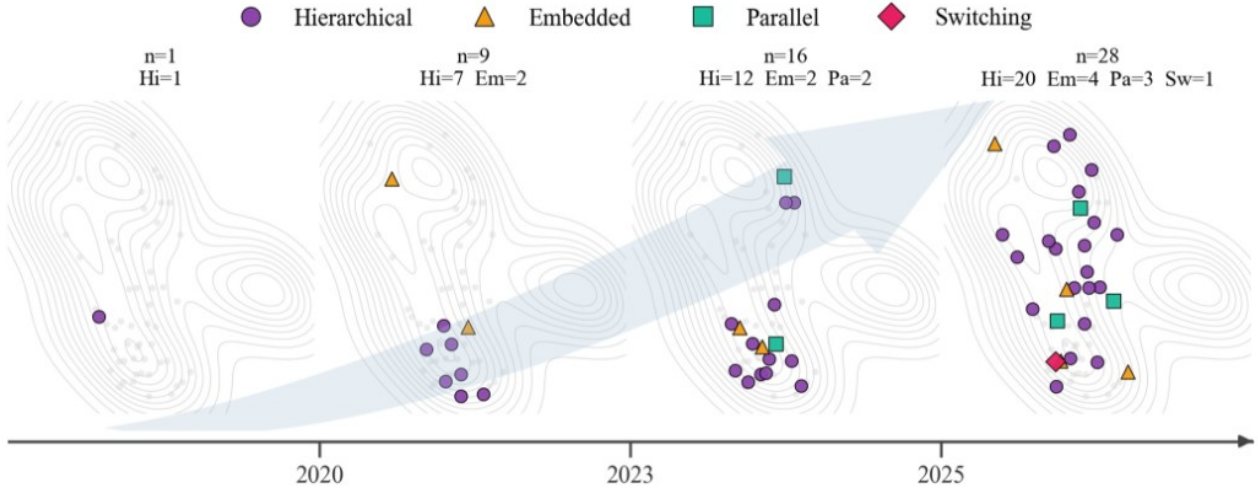


Figure 11: Temporal evolution of the 55 hybrid papers on the bibliometric UMAP, coloured by the four structural sub-classes (Hierarchical, Embedded, Parallel, Switching). Grey contours show the full WBC corpus; coloured markers are hybrids in each period. Pre-2020 literature is a single Hierarchical point; the later panels (2020–2022, 2023–2024, 2025+) show Hierarchical remaining dominant while Embedded, Parallel, and Switching appear as the field expands.

Table 7: Cross-category comparison of hybrid WBC structures. Ratings are qualitative (+ favourable, – unfavourable, ○ neutral) based on the surveyed literature.

Category	Sample efficiency	Safety / constraints	General-isation	Inter-pretability	Eng. complexity	Preferred deployment context
Hierarchical	+	○	+	○/+	+	General locomotion and loco-manipulation; first choice when a low-dim plan exists.
Switching	○	+	+	+	○	Multi-skill, long-horizon tasks where each skill has its own competent controller.
Parallel	+	+	○	+	○	Upgrading an existing model-based system with sensor-rich adaptation.
Embedded	○	+	○	+	–	High-precision tasks where a formal property of $\mathcal{M}$ must hold and learning must respect it.

a WBC stack—rather than as task wrappers—is an open architectural question. (v) *Sim-to-real for hybrid pipelines*. Each constituent already incurs a sim-to-real gap; their composition can either amplify or attenuate the gap depending on which side of the interface the mismatch lies on, and a systematic study of this is missing. (vi) *Sample- and data-efficient hybrids*. The surveyed embedded works typically inherit the data appetite of the underlying RL pipeline; whether differentiable $\mathcal{M}$ can be made to replace, rather than supplement, large simulated rollouts—through shorter horizons, latent dynamics, or analytical sensitivities—is not yet settled.

Looking across the six directions, the most concrete near-term agenda is methodological: a shared benchmark suite covering at least one locomotion task, one loco-manipulation task and one whole-

body manipulation task, each evaluated on more than one hybrid architecture per category, would do more to clarify the comparative landscape than any new architectural family. Until such a suite exists, the choice between hierarchical, switching, parallel and embedded designs will continue to be driven by team expertise and platform constraints, rather than by reproducible quantitative evidence.

6 Cross-Paradigm Synthesis: Knowledge, Objectives, Constraints, and Deployment

The three pillars each followed their own axis: model-based WBC the task-arbitration flexibility axis κ_{τ} (Sec. 3), learning-based WBC the data-diversity axis

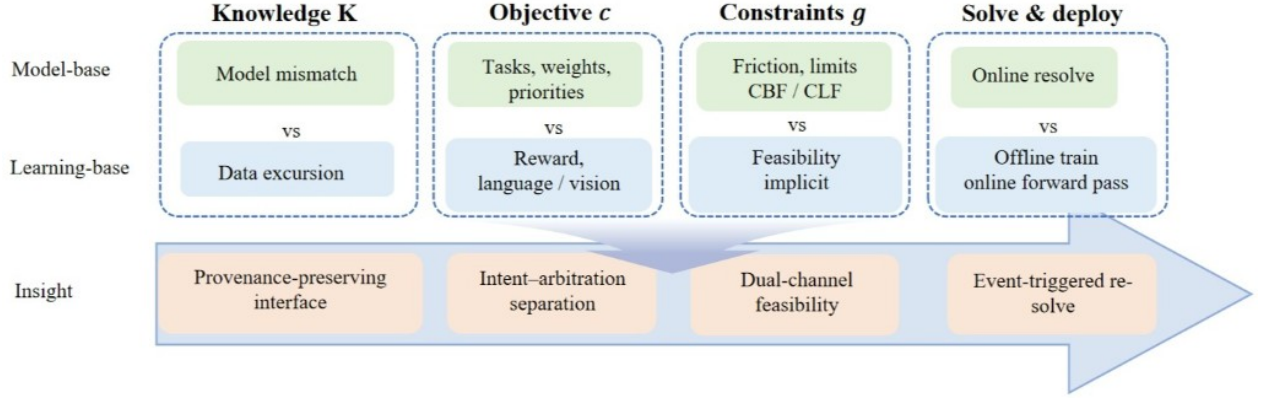


Figure 12: The three paradigms answer the same four design questions of (1). Each column contrasts the model-based and learning-based answers and closes on a hybrid insight: provenance-preserving interface ($\mathcal{K}$), intent-arbitration separation (c), dual-channel feasibility (g), and event-triggered re-solve (solve & deploy).

κ_d (Sec. 4), and hybrid WBC the coupling axis κ_{ML} (Sec. 5). This section adds no fourth taxonomy. It projects the three axes back onto the single problem they all solve.

Sec. 2 fixed the common language. Whether a controller is an equation, a network, or a composition of both, it computes (1): minimise a task cost c over admissible actions subject to a feasibility constraint g , both fed by a knowledge source $\mathcal{K}$. That expression is not an algorithm but a set of four design questions, namely where knowledge comes from, how objectives are defined and arbitrated, which conditions count as constraints, and when and by what solver the action is produced and deployed. The paradigms differ only in their answers. Model-based control writes $\mathcal{K}$, c , and g as explicit models and solves online; learning-based control compiles them from data into θ ; hybrid control decides which of them may be learned implicitly and which must remain an analytical scaffold at deployment.

Each subsection follows one order: recover the model-based answer, recover the learning-based answer, name the structural gap, and use a hybrid scaffold from Sec. 5 as evidence that the gap is fillable. Every subsection closes on one falsifiable insight, and Fig. 12 collects the four. **The chapter thesis is that hybrid WBC is not a fourth control problem but a redistribution, across $\mathcal{K}$, c , g , and the solver, of which design decisions stay explicit and which become learned.**

6.1 Knowledge: Model Fidelity versus Data Coverage

Model-based control grounds $\mathcal{K}$ in rigid-body dynamics, actuator models, contacts, and task kinematics; its value is less raw accuracy than traceability, since every term’s origin and validity stay visible in the solver. Its limit is the model-fidelity ceiling of Sec. 3.6: unknown contacts, compliant terrain, or payload shifts push $\mathcal{K}$ away from reality,

and a stronger solver then solves the wrong problem more precisely. Learning-based control widens $\mathcal{K}$ to rollouts, human motion, teleoperation, synthetic trajectories, and web-scale priors, absorbing regularities no equation captures; but data breadth is not feasibility coverage, and the source, confidence, and validity range of a learned reference are usually lost. The gap is thus explicit-but-local against broad-but-implicit, and the missing quantity is knowledge provenance and uncertainty, not data volume. Hierarchical scaffolds bridge it in one direction: 34 of the 55 surveyed hybrids let $\mathcal{M}$ produce feasible references while $\mathcal{L}$ supplies perception or intent through a narrow interface, yet the flow is almost entirely downward. *Insight 1 (knowledge interface)*. Model and data are complementary coordinates rather than substitutes; an interface that carries feasibility margin and uncertainty upward, not only setpoints downward, should lower catastrophic-failure rates under new payloads and contacts more than a wider dataset alone.

6.2 Objectives: Explicit Cost versus Learned Reward

Model-based control writes c as interpretable task errors, weights, priorities, and predictive costs; the κ_τ progression of Sec. 3 is exactly a tightening of how competing objectives are arbitrated, from soft weighting to lexicographic order to horizon-wide resolution. Its cost is the specification burden, since a designer must decide in advance how tracking, effort, smoothness, and contact trade off, and semantic goals rarely translate into a precise c . Learning-based control expresses objectives through reward, demonstrations, adversarial priors, and language or vision conditioning, discovering intent that is hard to hand-code; but it usually collapses task intent, priority, and soft constraints into one scalar return, so it cannot explain which objective was sacrificed in a conflict. The gap is that intent and arbitration are

conflated. The remedy splits c in two: learning fixes semantics, phase, and preference, while the analytical layer preserves priority and temporal trade-offs. Sec. 5 shows this directly, with learning setting cost weights, terminal values, or skill choices and the optimiser arbitrating them. *Insight 2 (intent–arbitration separation)*. Learning a bounded, low-dimensional objective interface and letting an HQP or MPC arbitrate it should preserve priorities under task switches and unseen goal combinations better than a direct action policy of equal training budget.

6.3 Constraints: Explicit g versus Implicit θ

Model-based control writes g as dynamics, contact consistency, friction cones, joint and torque limits, and CBF/CLF conditions; the solver reports which constraint is active and how much margin remains, so feasibility is enforced and diagnosable at once. Yet explicit is not correct: a mispredicted contact or a wrong learned geometry can exclude feasible actions or admit dangerous ones. Learning-based control instead acquires implicit feasibility from large-scale simulation and recovery training, staying balanced through contacts no explicit planner enumerated; but g dissolves into θ , failures surface only as falls, penalties bias behaviour in expectation without per-instant guarantees, and coverage does not survive distribution shift. The mirror is explicit-but-misspecified against adaptive-but-uncertified, and what is needed is two mutually checking channels, a learned estimate of hard-to-model feasibility and an analytical layer that enforces constraints and returns readable margins. Parallel and embedded safety layers realise this, though they remain the rarest classes, at 4 and 14 of 55 works. *Insight 3 (dual-channel feasibility)*. Under friction, load, or contact shift, pairing learned feasibility with an analytical safety layer should cut both the constraint-violation rate and the fault-localisation time, not merely raise mean success.

6.4 Solving and Deployment: Online Re-solving versus Forward Pass

Model-based control solves (1) online, recomputing the action every cycle and, with MPC, rescheduling conflicts over a horizon; this buys explicit adaptation and constraint enforcement at the price of solver latency that grows along κ_τ . Learning-based control moves the solve offline and runs a single forward pass, producing high-dimensional actions at low latency but unable to re-derive its problem once load, terrain, or contact leave the training envelope; history conditioning and randomisation interpolate, they do not re-solve. The gap is one of compute allocation rather than online versus offline, since a full solver is often too slow while a fixed policy stays too confidently wrong exactly when re-solving is needed.

Switching and adaptive scaffolds answer with event-triggered computation, running the fast policy in familiar regimes and invoking identification, short-horizon optimisation, residual correction, or a conservative fallback when risk rises. *Insight 4 (event-triggered re-solving)*. Triggering online computation on an out-of-distribution or constraint-margin signal should spend less average compute than always-on MPC and incur fewer out-of-envelope failures than a policy alone.

6.5 Synthesis: Selective Coupling

The four questions share one lesson. The taxonomy of Sec. 5 sorts each paper by its dominant topology, which a literature census requires; but a complete future stack may adopt a different scaffold at each stage, generating training data through a hierarchical $\mathcal{M} \rightarrow \mathcal{L}$ path, setting goals through a semantic module, enforcing constraints through an embedded safety layer, and switching to adaptive MPC out of distribution. Topology answers what a paper’s main structure is; this section answers where the analytical scaffold belongs in the pipeline. It follows that κ_{ML} need not increase monotonically across a system. Deeper coupling buys expressiveness at the cost of engineering complexity and training brittleness, so the design goal is to tighten coupling locally where an error is most costly and keep the rest modular. **Selective-coupling principle: hybrid WBC should retain the minimal analytical scaffold exactly where data is least trustworthy, objectives most ambiguous, constraints most consequential, or deployment most in need of re-solving, and learn everything else.** Read this way, (1) closes: $\mathcal{K}$ sets what the controller believes, c what it pursues, g what it must never do, and the solver when it may change its answer. The three paradigms are not closed worlds but different allocations of explicit structure and learned capacity across these four decisions, and hybrid control is the transition in which that allocation is, for the first time, chosen deliberately.

Table 8 turns this principle into a reader’s map. Each row starts from a bottleneck already named in Secs. 3–5, names the research direction that would loosen it, and points to the architecture and deployment setting in which that direction is most likely to pay off. The table is a starting guide rather than a ranking: identify the bottleneck that dominates a new problem, then read across.

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Table 8: A reader’s map from bottleneck to research direction, architecture, and deployment setting. Architectures follow the taxonomy of Sec. 5.

Bottleneck	Research direction	Suitable architecture	Suitable scenario
Model fidelity versus data coverage: explicit $\mathcal{K}$ is local, learned $\mathcal{K}$ is broad but loses provenance	Provenance-preserving interface; carry feasibility margin and uncertainty upward, not only setpoints downward	Hierarchical	General locomotion and loco-manipulation when a low-dimensional plan or reference already exists
Specification burden versus collapsed intent: c is either hand-coded and brittle or a single scalar return	Intent–arbitration separation: learn semantics and preference, keep analytical priority	Hierarchical (skill or cost modulation); Embedded (learn the cost)	Language- or vision-specified goals; frequent task switches and unseen goal combinations
Explicit-but-misspecified g versus implicit-but-uncertified θ : faults are either wrong constraints or opaque falls	Dual-channel feasibility: a learned estimate checked by an analytical safety layer	Parallel residual; Embedded safety	High-precision or safety-critical contact, payload, or friction shift
Always-on solve too slow versus a frozen policy that cannot re-derive its problem out of distribution	Event-triggered re-solving on an out-of-distribution or constraint-margin signal	Switching; adaptive residual MPC	Multi-skill or long-horizon tasks; regimes that leave the training envelope

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